

Reservoir Computing in Wireless Receive Processing: An Information-Theoretic Perspective

Shashank Jere¹, Lizhong Zheng², *Fellow, IEEE*, Ummay Sumaya Khan¹, and Lingjia Liu², *Fellow, IEEE*

Abstract—Deep learning is playing a transformational role in the physical layer of wireless communications. While demonstrating remarkable empirical performance in applications such as MIMO detection and receive processing, the fundamental reasons underlying this superlative performance remain insufficiently understood. In this work, we advance the development of Explainable AI (xAI) within the physical layer by grounding the analysis of deep learning methods in estimation-theoretic and information-theoretic principles. Specifically, we focus on wireless receive processing, namely channel equalization through reservoir computing (RC)—a recurrent neural network (RNN) framework that has shown strong empirical receive processing performance in both Orthogonal Frequency Division Multiplexing (OFDM) and Orthogonal Time Frequency Space (OTFS) systems. Within RC, we focus on the echo state network (ESN) and its extension, the windowed ESN (WESN). Building on our prior work that establishes the signal processing foundations of the ESN in receive processing, we conduct an exact analytical characterization of the equalization performance of the ESN and WESN and compare them against the optimum Wiener equalizer in a wireless fading channel with inter-symbol interference (ISI). The validity of the theoretical results is established through extensive numerical evaluations.

Index Terms—Deep learning, recurrent neural networks, explainable machine learning, model interpretability, reservoir computing, echo state network, equalization, receive processing, symbol detection, mutual information.

I. INTRODUCTION

THE recent surge of deep learning has been remarkable, driven largely by its outstanding empirical success across a wide range of applications. Wireless communications have rapidly followed this trend, with machine learning (ML) and neural network (NN) methods increasingly applied at the physical layer. Artificial intelligence (AI)-enabled networks are expected to play a pivotal role in meeting the stringent demands of next-generation (NextG) wireless systems [1], [2]. These demands are compounded by challenges such as network complexity, model deficit, and algorithm deficit [1], which constrain the applicability of conventional model-based approaches for physical layer (PHY) processing, including

MIMO detection. By contrast, AI-driven methods for PHY processing have emerged as a promising alternative to overcome these limitations [2].

Most current deep learning strategies rely on training large neural networks offline with extensive datasets, followed by deployment for inference. However, this approach is often impractical in wireless PHY tasks, where over-the-air (OTA) training data is scarce. A further obstacle is the well-known uncertainty in generalization [1], which arises from mismatches between offline training and online deployment in system configurations, channel conditions, and operational dynamics [2]. Consequently, purely offline learning is unsuitable for data-limited, low-latency PHY applications. Online and hybrid learning approaches have therefore been proposed as viable alternatives. Within this context, reservoir computing (RC) [3] provides efficient online real-time learning due to its low-complexity training procedure. This makes RC highly suited for complexity- and latency-constrained PHY tasks, such as MIMO-OFDM detection [4], [5], [6] and dynamic spectrum access [7], [8], where it has shown superior performance compared to traditional model-based and offline learning approaches [9], [10].

Recent advances in deep neural network (DNN)-based MIMO symbol detection have also attracted significant attention. Approaches such as *DetNet* [11], *MMNet* [9], *OAMPNet* [12], and *HyperMIMO* [13] incorporate learnable parameters into iterative detection algorithms and achieve strong performance. However, these methods require substantial training data and generally assume perfect channel state information (CSI), which is difficult or impossible to obtain in practice. By contrast, RC-based techniques, with their lightweight training requirements, present an attractive alternative. Early applications of echo state networks (ESNs) for MIMO-OFDM detection were introduced in [14]. Subsequent enhancements, such as windowed inputs, skip connections, and delayed outputs, led to the WESN architecture [4], which demonstrated significant improvements over the vanilla ESN. Further developments include the deep RC structure *RCNet* [5] and *RC-Struct* [6], which exploit the time-frequency structure of OFDM signals. Our previous work also include application of the RC architecture in a way that exploits the delay-Doppler domain channel representation to improve OTFS symbol detection performance [15]. In other works, ESNs have also been shown to be effective for nonlinear channel equalization [16], especially when employed in conjunction with Kalman filtering techniques [17]. These methods have shown consider-

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Shashank Jere, Ummay Sumaya Khan, and Lingjia Liu are with Wireless@VT, Bradley Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA 24061 USA (e-mail: ljliu@ieee.org).

Lizhong Zheng is with the EECS Department, Massachusetts Institute of Technology, Cambridge, MA 02139 USA.

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able gains over both traditional signal processing approaches and other learning-based methods. A key advantage of RC methods is that training can be conducted fully online, on a slot-by-slot basis, with significantly reduced complexity. This allows RC-based detectors to adapt effectively to dynamic transmission modes in modern networks, retraining in real time using only the limited OTA reference signals available. Despite these strong empirical results, the general effectiveness and interpretability of RC-based methods for PHY receive processing have not yet been fully understood.

In parallel with the adoption of deep learning, there has been a growing emphasis on the explainability of such methods, giving rise to the fields of explainable AI (xAI) and explainable machine learning (xML) [18]. Early theoretical analyses of RC include the functional space approximation framework of [19] and the demonstration in [20] that linear ESNs are equivalent to vector autoregression (VAR). RC's ability to model nonlinear dynamical systems such as the Lorenz and Rössler attractors has been explored in [21], while memory length optimization was studied in [22]. Our earlier work [23] derived an upper bound on the empirical Rademacher complexity of single-reservoir ESNs, showing they achieve tighter generalization than conventional RNNs and illustrating how this bound can be used to optimize ESN-based MIMO-OFDM symbol detectors. Related research, such as [24], has also established generalization error bounds for RC using modified Rademacher-type complexity measures. In our prior work [25], we offered a signal processing perspective on ESNs and analytically characterized the optimal untrained recurrent weight for single-neuron ESNs in channel equalization tasks.

This paper builds upon our previous work in explainable artificial intelligence (xAI), specifically the application of RC/ESN in wireless receive processing, including MIMO-OFDM symbol detection [26], [27], [28]. In this work, we present theoretical results that quantify the performance achieved by the ESN as an equalizer in receive processing, and benchmark its performance against optimum solutions. Furthermore, we also theoretically justify the enhanced performance of the windowed ESN (WESN) in receive processing applications seen empirically and precisely quantify its performance gain over the vanilla ESN. For both the ESN and the WESN, which form the core of many sophisticated RC-based symbol detectors [4], [5], [6], we clearly quantify their respective gaps from optimality using standard estimation theoretic measures. Furthermore, for both architectures under linear activation, we provide an alternative information theoretic viewpoint which arrives at the same conclusions, further strengthening the validity of the overall analysis.

The main contributions of this work can be summarized as follows:

- Under a two-tap fading channel with inter-symbol interference (ISI), we provide an exact characterization of the minimum mean squared error (MMSE) achieved by the optimum Wiener equalizer, the vanilla echo state network (ESN), the windowed ESN (WESN) as well as an FIR-only linear MMSE (LMMSE) equalizer.
- For a given fading channel realization, we rigorously establish that the WESN achieves a strictly lower post-

equalization MSE than the vanilla ESN, both initialized with the same set of randomly generated reservoir recurrent weights or equivalently, reservoir poles.

- A useful geometric interpretation of the WESN equalization gain is provided as an orthogonal decomposition using two representations: i) ESN-centric, and ii) FIR-centric. These dual but equivalent representations provide an intuitive understanding of the superior equalization performance exhibited by the WESN.
- An alternative information-theoretic perspective on the equalization performance of each receive processing method is provided leveraging the linkages between mutual information and MMSE under Gaussian signaling [29], reaching the same MSE results as those derived using exact estimation-theoretic analysis.
- The derived theoretical results on the MSE performance of each receive processing method are thoroughly validated with numerical evaluations.

Notation: $\mathbf{a}$ denotes a column vector. $\mathbf{A}$ and $\mathbf{A}$ denote matrices. $[\mathbf{A}]_{i,j}$ denotes the (i, j) -th element of $\mathbf{A}$. $[\cdot, \cdot]$ denotes horizontal concatenation and $[\cdot; \cdot]$ denotes vertical concatenation respectively of scalars, vectors or matrices. $(\cdot)^*$ denotes the complex conjugate. $\mathbf{0}_{M \times N}$ denotes the $M \times N$ all-zeros matrix. $\mathbf{I}_N$ denotes the $N \times N$ identity matrix. $(\cdot)^H$ denotes the matrix conjugate transpose. $(\cdot)^\dagger$ denotes the Moore-Penrose matrix pseudoinverse. $\text{diag}(\cdot)$ denotes a diagonal matrix formed by scalar arguments; $\text{tridiag}(\cdot; \cdot; \cdot)$ denotes a tridiagonal Toeplitz matrix with the respective subdiagonal, main diagonal and superdiagonal entries.

The remainder of this paper is organized as follows. Sec. II introduces the system model that forms the basis of the analysis in this work. Sec. III presents an estimation-theoretic analysis and derivation of the exact post-equalization MSE and SINR expressions for each considered linear equalizer. An information-theoretic perspective of the equalization performance and its implications on mutual information and MMSE is provided in Sec. IV. The theoretical results are validated through empirical numerical evaluations in Sec. V. Finally, Sec. VI concludes the paper and provides an outlook for future work.

II. SYSTEM MODEL

In this work, we consider a two-tap wireless fading channel with the channel impulse response (CIR) given by $[1, -\alpha]$, where $\alpha \in \mathbb{C}$ and $u \triangleq |\alpha|^2$. Denoting the transmit symbol stochastic process as $\{x_n\}$ and the additive Gaussian noise stochastic process as $\{w_n\}$ respectively, the received symbol stochastic process $\{y_n\}$ can be expressed as

$$y_n = x_n - \alpha x_{n-1} + w_n. \quad (1)$$

We make the following assumptions:

Assumption 1: In the derivations that follow and unless otherwise specified, the channel fading coefficient α is assumed to be deterministic. Alternatively, the derivations can be regarded as valid for a specific realization α of the fading coefficient random variable $\mathbf{A}$. Therefore, the expectation operator is applied over the joint distribution of x_n and w_n , i.e., $\mathbb{E}_{x,w}[\cdot]$ is henceforth abbreviated as $\mathbb{E}[\cdot]$, unless otherwise specified.

Assumption 2: $\{x_n\}$ and $\{w_n\}$ are mutually independent, zero-mean, proper-complex and temporally white Gaussian stochastic processes, i.e., given $x_n \sim \mathcal{CN}(0, P)$, $w_n \sim \mathcal{CN}(0, N_0)$, we have $\mathbb{E}[x_n x_m^*] = P\delta[n - m]$ and $\mathbb{E}[w_n w_m^*] = N_0\delta[n - m]$ and $\mathbb{E}[x_n w_m^*] = 0 \quad \forall n, m \in \mathbb{Z}$.

The two-tap channel therefore induces intersymbol interference (ISI) and has the frequency response represented by its discrete-time Fourier transform (DTFT) as $H(e^{j\omega}) = 1 - \alpha e^{j\omega}$. The nominal signal-to-noise ratio (SNR) is defined as $\gamma \triangleq P/N_0$. Integration in ω is over $[-\pi, \pi]$.

III. ESTIMATION-THEORETIC BENCHMARKING OF LINEAR EQUALIZERS

In this section, we provide performance results of multiple linear estimator classes for the received signal model introduced in Sec. II. Specifically, we precisely calculate the post-equalization mean square error (MSE) and the signal-to-noise-plus-interference ratio (SINR) for every linear estimator class. For a given LTI equalization filter G that acts on $\{y_n\}$, define the estimation error as $e_n \triangleq x_n - \hat{x}_n$ and the corresponding error spectrum as $S_{ee}(\omega)$. Then, the mean square error is

$$\text{MSE}(G) = \mathbb{E}[|x_n - \hat{x}_n|^2] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{ee}(\omega) d\omega, \quad (2)$$

where $\hat{x}_n = \sum_{k \in \mathcal{K}} g_k y_{n-k}$, where $\mathcal{K}$ denotes the set of indices of the received samples $\{y_n\}$ that the particular equalizer G uses.

Subsequently, we study the following classes of the equalization/estimation filter G :

- The globally optimum non-causal Wiener equalizer.
- A vanilla ESN with M reservoir neurons under linear activation.
- A windowed ESN (WESN) [4] with M reservoir neurons and output skip (FIR) connections of length N_w under linear activation.
- A N_w -tap causal FIR LMMSE equalizer.

A. Globally Optimum Wiener Equalizer (Non-Causal)

The Wiener solution that achieves the minimum mean square error (MMSE) is given by

$$G_{\text{opt}} = \arg \min_G \text{MSE}(G) \quad (3)$$

We derive the frequency response $G_{\text{opt}}(e^{j\omega})$ of the global optimum Wiener equalizer and the corresponding post-equalization MSE_{opt} and SINR_{opt} by first stating the following lemma.

Lemma 1: For the two-tap ISI channel described in Eq. (1), the power spectral densities of $\{x_n\}$, $\{y_n\}$ and the cross spectral density of $\{(x_n, y_n)\}$ are given by $S_{xx}(\omega) = P$, $S_{yy}(\omega) = P|H(e^{j\omega})|^2 + N_0$ and $S_{xy}(\omega) = PH^*(e^{j\omega})$.

Proof: See Appendix A-A. $\square$

Using tools from Wiener-Kolmogorov theory and by invoking the orthogonality principle, we can obtain the frequency response of the optimal Wiener equalizer $G_{\text{opt}}(e^{j\omega})$ and the corresponding error spectrum $S_{ee}(\omega)$. The result is stated in the below lemma.

Lemma 2: The MMSE-achieving Wiener equalizer has the frequency response given by

$$G_{\text{opt}}(e^{j\omega}) = \frac{S_{xy}(\omega)}{S_{yy}(\omega)} = \frac{PH^*(e^{j\omega})}{P|H(e^{j\omega})|^2 + N_0}, \quad (4)$$

and induces the error spectrum given by

$$S_{ee}(\omega) = S_{xx}(\omega) - \frac{|S_{xy}(\omega)|^2}{S_{yy}(\omega)} = P - \frac{P^2|H(e^{j\omega})|^2}{P|H(e^{j\omega})|^2 + N_0}. \quad (5)$$

Proof: See Appendix A-B. $\square$

The minimum MSE achieved by the (non-causal) Wiener equalizer can now be evaluated in closed form as

$$\text{MSE}_{\text{opt}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{ee}(\omega) d\omega = \frac{PN_0}{2\pi} \int_{-\pi}^{\pi} \frac{d\omega}{P|H(e^{j\omega})|^2 + N_0}. \quad (6)$$

We can observe that

$$|H(e^{j\omega})|^2 = 1 + u - 2|\alpha| \cos(\omega - \arg \alpha), \quad (7)$$

where $u \triangleq |\alpha|^2$. Let $\theta \triangleq \omega - \arg \alpha$ and define the positive scalars $A \triangleq P(1 + u) + N_0$ and $B \triangleq 2P\sqrt{u}$. Then, Eq. (6) becomes

$$\text{MSE}_{\text{opt}} = \frac{PN_0}{2\pi} \int_{-\pi}^{\pi} \frac{d\theta}{A - B \cos \theta}. \quad (8)$$

The above trigonometric integral admits a closed-form solution and it can be shown that

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d\theta}{A - B \cos \theta} = \frac{1}{\sqrt{A^2 - B^2}}, \quad (9)$$

provided $A > |B|$ (see Appendix A-C for the proof). With all of the above results, we obtain the theorem below.

Theorem 1: For the two-tap ISI channel described in Eq. (1), the globally optimum (non-causal) Wiener equalizer achieves

$$\begin{aligned} \text{MSE}_{\text{opt}} &= \frac{PN_0}{\sqrt{(P(1 + u) + N_0)^2 - 4P^2u}} \\ &= \frac{P}{\gamma \sqrt{\left(1 + u + \frac{1}{\gamma}\right)^2 - 4u}}. \end{aligned} \quad (10)$$

Proof: The above result follows directly from substituting the integral solution in Eq. (9). $\square$

Corollary 1: The post-equalization SINR of the globally optimum (non-causal) Wiener equalizer is

$$\text{SINR}_{\text{opt}} = \gamma \sqrt{\left(1 + u + \frac{1}{\gamma}\right)^2 - 4u} - 1. \quad (11)$$

Proof: For the MMSE estimator, which is also the LMMSE estimator due to the joint Gaussianity of x_n and y_n , the orthogonality principle states that the error e_n is orthogonal to the estimate $\hat{x}_n$, i.e., $\mathbb{E}[\hat{x}_n e_n] = 0$. It follows that

$$\begin{aligned} \mathbb{E}[|x_n|^2] &= \mathbb{E}[|\hat{x}_n|^2] + \mathbb{E}[|e_n|^2], \\ \Rightarrow \mathbb{E}[|\hat{x}_n|^2] &= P - \text{MSE}_{\text{opt}}. \end{aligned} \quad (12)$$

Following the definition of the post-equalization SINR, we have

$$\text{SINR}_{\text{opt}} = \frac{\mathbb{E}[|\hat{x}_n|^2]}{\mathbb{E}[|e_n|^2]} = \frac{P - \text{MSE}_{\text{opt}}}{\text{MSE}_{\text{opt}}} = \frac{P}{\text{MSE}_{\text{opt}}} - 1. \quad (13)$$

Substituting Eq. (10) for MSE_{opt} obtained in Theorem 1, we get the required expression for the SINR of the optimum non-causal Wiener equalizer in Eq. (11). $\square$

Remark 1: Both MSE_{opt} and SINR_{opt} exhibit phase invariance w.r.t. the channel fading coefficient α , i.e., they only depend on the tap power $u = |\alpha|^2$.

Remark 2: For an AWGN channel with no ISI, i.e., $\alpha = 0 \Rightarrow u = 0$, we get $\text{MSE}_{\text{opt}} = P/(1 + \gamma)$ and $\text{SINR}_{\text{opt}} = \gamma$. This aligns exactly with the well-established LMMSE result for a scalar AWGN channel. Furthermore, as the nominal SNR $\gamma \rightarrow \infty$ with no ISI, we get $\text{MSE}_{\text{opt}} \rightarrow 0$ and $\text{SINR}_{\text{opt}} \rightarrow \infty$, as expected.

Remark 3: (High-SNR Asymptotics) Consider $\gamma \rightarrow \infty$ for $u \neq 0$. We have two cases here:

- If $u \neq 1$, then

$$\begin{aligned} \text{MSE}_{\text{opt}} &= \frac{P}{\gamma} \cdot \frac{1}{\sqrt{(1+u+1/\gamma)^2 - 4u}} \sim \frac{P}{\gamma|1-u|} \\ &= \frac{N_0}{|1-u|}. \end{aligned} \quad (14)$$

Therefore, the MSE decays as N_0 for $u \neq 1$.

- If $u = 1$, $A^2 - B^2 = (2P + N_0)^2 - 4P^2 = 4PN_0 + N_0^2$, then

$$\text{MSE}_{\text{opt}} = \frac{PN_0}{\sqrt{N_0(4P + N_0)}} \sim \frac{1}{2} \sqrt{PN_0}. \quad (15)$$

Therefore, the MSE decays as $\sqrt{N_0}$ when the channel zero lies on the unit circle ($u = 1$), indicating a much slower rate of MSE reduction for an “unfavorably” placed channel zero.

B. Vanilla ESN With M Neurons

We consider a “vanilla” echo state network (ESN) with linear activation and M non-interconnected neurons in its reservoir. The qualifier “vanilla” here signifies an architecture consisting only of a single reservoir with associated fixed input weights equal to unity. The corresponding recurrent weights of the neurons, which are also poles of a first-order IIR filter are denoted by $\beta_i \in \mathbb{C}$ satisfying $|\beta_i| < 1$ for $i = 1, 2, \dots, M$. The state update equation for the i^{th} neuron at discrete-time index n can be written as

$$r_{n,i} = \beta_i r_{n-1,i} + y_n, \quad i = 1, 2, \dots, M. \quad (16)$$

Here, $r_{n,i} \in \mathbb{C}$ denotes the state of the i -th neuron at discrete time index n and y_n denotes the input to the ESN at time index n . Note that y_n is the channel output from Eq. (1). The state vector is $\mathbf{r}_n = [r_{n,1}, r_{n,2}, \dots, r_{n,M}]^T \in \mathbb{C}^M$. The equalized symbol is produced by a linear output (readout) layer $\mathbf{w}_{\text{out}} \in \mathbb{C}^M$ acting on $\mathbf{r}_n$ as

$$\hat{x}_n = \mathbf{w}_{\text{out}}^H \mathbf{r}_n, \quad (17)$$

where $\hat{x}_n \in \mathbb{C}$ is the estimate of the transmitted symbol x_n and $\mathbf{w}_{\text{out}}$ is chosen to minimize $\mathbb{E}[|x_n - \hat{x}_n|^2]$. With perfect training and ideal knowledge of the second-order statistics of the underlying WSS processes, the optimum readout weights $\mathbf{w}_{\text{out,opt}}$ can be obtained via the Normal equations as

$$\mathbf{w}_{\text{out,opt}} = \mathbf{R}_{rr}^{-1} \mathbf{r}_{rx}, \quad (18)$$

where $\mathbf{R}_{rr} \triangleq \mathbb{E}[\mathbf{r}_n \mathbf{r}_n^H] \in \mathbb{C}^{M \times M}$ is the reservoir covariance matrix and $\mathbf{r}_{rx} \triangleq \mathbb{E}[\mathbf{r}_n x_n^*] \in \mathbb{C}^M$ is the reservoir-transmit symbol cross-covariance vector. Invoking the orthogonality principle, it follows that

$$\text{MSE}_{\text{ESN}} = P - \mathbf{r}_{rx}^H \mathbf{R}_{rr}^{-1} \mathbf{r}_{rx}, \quad (19)$$

$$\text{SINR}_{\text{ESN}} = \frac{P}{\text{MSE}_{\text{ESN}}} - 1. \quad (20)$$

Before deriving closed-form expressions for the above mentioned second-order statistics, we first derive some useful relationships involving the autocorrelation $R_{yy}[\ell] \triangleq \mathbb{E}[y_n y_{n-\ell}^*]$. For the two-tap ISI channel of Eq. (1) and under Assumption 1 and 2, it follows that

$$\begin{aligned} R_{yy}[0] &= \mathbb{E}[|y_n|^2] \\ &= \mathbb{E}[|x_n|^2] + |\alpha|^2 \mathbb{E}[|x_{n-1}|^2] + \mathbb{E}[|w_n|^2] \\ &= P(1 + u) + N_0, \end{aligned} \quad (21)$$

$$\begin{aligned} R_{yy}[1] &= \mathbb{E}[y_n y_{n-1}^*] \\ &= \mathbb{E}[(-\alpha x_{n-1})(x_{n-1})^*] = -\alpha \mathbb{E}[|x_{n-1}|^2] \\ &= -\alpha P, \end{aligned} \quad (22)$$

$$R_{yy}[-1] = (R_{yy}[1])^* = -\alpha^* P, \quad (23)$$

$$R_{yy}[\ell] = 0, \quad |\ell| \geq 2. \quad (24)$$

Let us define the scalars $R_0 \triangleq R_{yy}[0]$, $R_1 \triangleq R_{yy}[1]$, $R_{-1} \triangleq R_{yy}[-1]$. Furthermore, define $\mathbf{v} \triangleq [\beta_1, \beta_2, \dots, \beta_M]^T \in \mathbb{C}^M$ and $\mathbf{w} \triangleq [\beta_1^2, \beta_2^2, \dots, \beta_M^2]^T \in \mathbb{C}^M$. Equipped with the above, we derive closed-form expressions for $\mathbf{r}_{rx}$ and $\mathbf{R}_{rr}$ in the following two lemmas.

Lemma 3: The reservoir-transmit cross-covariance is given by $\mathbf{r}_{rx} = P \mathbf{1}_M$.

Proof: We can rewrite the reservoir state for the i^{th} neuron as $r_{n,i} = \sum_{k \geq 0} \beta_i^k y_{n-k}$. Also, due to the independence of x_n from past symbols and noise, we have $\mathbb{E}[y_{n-k} x_n^*] = 0$ for $k \geq 1$. Furthermore, $\mathbb{E}[y_n x_n^*] = P$. Therefore, we arrive at $\mathbb{E}[r_{n,i} x_n^*] = P$, resulting in $\mathbf{r}_{rx} = \mathbb{E}[\mathbf{r}_n x_n^*] = P \mathbf{1}_M$. $\square$

Lemma 4: For $|\beta_i| < 1$, the (i, j) -th element of $\mathbf{R}_{rr}$ is given by

$$[\mathbf{R}_{rr}]_{i,j} = \frac{R_0 + R_1 \beta_j^* + R_{-1} \beta_i}{1 - \beta_i \beta_j^*}, \quad 1 \leq i, j \leq M. \quad (25)$$

Proof: Using $r_{n,i} = \sum_{k \geq 0} \beta_i^k y_{n-k}$ and defining $R_{yy}[\ell] \triangleq \mathbb{E}[y_n y_{n-\ell}^*]$, we get $R_{yy}[0] = R_0$, $R_{yy}[1] = R_1$, $R_{yy}[-1] = R_{-1}$ and $R_{yy}[\ell] = 0$ for $\ell \geq 2$. Then, it follows from definition that

$$\begin{aligned} [\mathbf{R}_{rr}]_{i,j} &= \sum_{k, \ell \geq 0} \beta_i^k (\beta_j^*)^\ell R_{yy}[\ell - k], \\ &= \sum_{k \geq 0} \beta_i^k (\beta_j^*)^k R_0 + \sum_{k \geq 0} \beta_i^k (\beta_j^*)^{k+1} R_1 + \sum_{k \geq 1} \beta_i^k (\beta_j^*)^{k-1} R_{-1}, \\ &= \frac{R_0}{1 - \beta_i \beta_j^*} + \frac{R_1 \beta_j^*}{1 - \beta_i \beta_j^*} + \frac{R_{-1} \beta_i}{1 - \beta_i \beta_j^*}. \end{aligned} \quad (26)$$

Due to the symmetry of the constituent terms in Eq. (26), it follows that $\mathbf{R}_{rr}$ is Hermitian symmetric and positive definite for $N_0 > 0$. Therefore, $\mathbf{R}_{rr}$ is invertible. $\square$

Finally, substituting the results of Lemmas 3 and 4 in Eq. (19), we arrive at

$$\text{MSE}_{\text{ESN}} = P - P^2 \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M. \quad (27)$$

Eq. (27) above is exact for every realization of the channel fading coefficient α and for every set of *deterministic* ESN reservoir poles $\{\beta_i\}_{i=1}^M$ with $|\beta_i| < 1$. In what follows, we develop a rank-2 resummed form of the scalar $\mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M$, resulting in a much simpler expression that provides crucial insights and enables efficient evaluation.

Rank-reduced MSE Expression for the vanilla ESN:

Define $\mathbf{A} \triangleq [\mathbf{1}_M \quad \mathbf{v}] \in \mathbb{C}^{M \times 2}$ and $\mathbf{R}_{zz} \triangleq \begin{bmatrix} R_0 & R_1 \\ R_{-1} & R_0 \end{bmatrix} \in \mathbb{C}^{2 \times 2}$. Introduce the tail matrix $\mathbf{C}_{\text{tail}} \in \mathbb{C}^{M \times M}$ whose (i, j) -th entry ($1 \leq i, j \leq M$) is

$$[\mathbf{C}_{\text{tail}}]_{i,j} = \frac{(\beta_i \beta_j^*)^2}{1 - \beta_i \beta_j^*} (R_0 + R_1 \beta_j^* + R_{-1} \beta_i). \quad (28)$$

Set the *base matrix* as

$$\mathbf{R}_0 \triangleq \mathbf{A} \mathbf{R}_{zz} \mathbf{A}^H + \mathbf{C}_{\text{tail}} \in \mathbb{C}^{M \times M}. \quad (29)$$

Finally, define $\mathbf{U} \triangleq [\mathbf{v} \quad \mathbf{w}] \in \mathbb{C}^{M \times 2}$ and

$$\mathbf{C} \triangleq \begin{bmatrix} 0 & R_1 \\ R_{-1} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\alpha P \\ -\alpha^* P & 0 \end{bmatrix} \in \mathbb{C}^{2 \times 2}. \quad (30)$$

We are now ready to state the following proposition.

Proposition 1: The reservoir covariance matrix $\mathbf{R}_{rr}$ admits the exact rank-2 decomposition given by $\mathbf{R}_{rr} = \mathbf{R}_0 + \mathbf{U} \mathbf{C} \mathbf{U}^H$.

Proof: See Appendix B-A. $\square$

Following the decomposition of $\mathbf{R}_{rr}$ from Proposition 1, Woodbury's matrix inverse identity gives

$$\mathbf{R}_{rr}^{-1} = \mathbf{R}_0^{-1} - \mathbf{R}_0^{-1} \mathbf{U} (\mathbf{C}^{-1} + \mathbf{U}^H \mathbf{R}_0^{-1} \mathbf{U})^{-1} \mathbf{U}^H \mathbf{R}_0^{-1}, \quad (31)$$

provided $\mathbf{C}$ is non-singular ($\det(\mathbf{C}) \neq 0$), which holds for $\alpha \neq 0$ (the degenerate case of $\alpha = 0$ is trivial resulting in $\mathbf{C} = \mathbf{0}$). Therefore,

$$\mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M = \mathbf{1}_M^H \mathbf{R}_0^{-1} \mathbf{1}_M - \mathbf{b}^H (\mathbf{C}^{-1} + \mathbf{G})^{-1} \mathbf{b}, \quad (32)$$

where $\mathbf{b} \triangleq \mathbf{U}^H \mathbf{R}_0^{-1} \mathbf{1}_M = \begin{bmatrix} \mathbf{v}^H \mathbf{R}_0^{-1} \mathbf{1}_M \\ \mathbf{w}^H \mathbf{R}_0^{-1} \mathbf{1}_M \end{bmatrix}$ and $\mathbf{G} \triangleq \mathbf{U}^H \mathbf{R}_0^{-1} \mathbf{U} =$

$\begin{bmatrix} \mathbf{v}^H \mathbf{R}_0^{-1} \mathbf{v} & \mathbf{v}^H \mathbf{R}_0^{-1} \mathbf{w} \\ \mathbf{w}^H \mathbf{R}_0^{-1} \mathbf{v} & \mathbf{w}^H \mathbf{R}_0^{-1} \mathbf{w} \end{bmatrix}$. Defining $\mathbf{M} \triangleq (\mathbf{C}^{-1} + \mathbf{G}) \in \mathbb{C}^{2 \times 2}$ and $\xi_{\text{ESN}} \triangleq \mathbf{1}_M^H \mathbf{R}_0^{-1} \mathbf{1}_M - \mathbf{b}^H \mathbf{M}^{-1} \mathbf{b}$, we arrive at the following compact rank-reduced expression

$$\text{MSE}_{\text{ESN}} = P - P^2 \xi_{\text{ESN}}. \quad (33)$$

C. Windowed ESN (WESN) With M Reservoir Neurons and N_w FIR Window Taps

The linear WESN consists of a linear, non-interconnected reservoir of M neurons with poles $\beta_i \in \mathbb{C} (|\beta_i| < 1)$ and an N_w -sample window over received samples that implements an FIR filter of order N_w . The state update equation is the same as the vanilla ESN, i.e., $r_{n,i} = \beta_i r_{n-1,i} + y_n$, with the reservoir state vector $\mathbf{r}_n \in \mathbb{C}^M$ being similarly defined. For the WESN, we define the windowed observation vector of the input samples as $\mathbf{y}_n^{(w)} \triangleq [y_n, y_{n-1}, \dots, y_{n-N_w+1}]^T \in \mathbb{C}^{N_w}$. The observation/feature vector for the WESN is

$$\mathbf{s}_n = \begin{bmatrix} \mathbf{r}_n \\ \mathbf{y}_n^{(w)} \end{bmatrix} \in \mathbb{C}^{M+N_w}, \quad (34)$$

and the equalized symbol output by the WESN is

$$\hat{x}_n = \mathbf{w}_{\text{out}}^H \mathbf{s}_n, \quad (35)$$

where $\mathbf{w}_{\text{out}} \in \mathbb{C}^{M+N_w}$ are the output (readout) weights, now also including weights for the windowed FIR taps. The additional second-order quantities needed to find the optimum readout weights $\mathbf{w}_{\text{out,opt}}$ for the WESN via the Normal equations are

$$\mathbf{R}_{yy}^{(w)} \triangleq \mathbb{E} [\mathbf{y}_n^{(w)} (\mathbf{y}_n^{(w)})^H] \in \mathbb{C}^{N_w \times N_w}, \quad (36)$$

$$\mathbf{R}_{ry} \triangleq \mathbb{E} [\mathbf{r}_n (\mathbf{y}_n^{(w)})^H] \in \mathbb{C}^{M \times N_w}, \quad (37)$$

$$\mathbf{r}_{yx}^{(w)} \triangleq \mathbb{E} [\mathbf{y}_n^{(w)} x_n^*] \in \mathbb{C}^{N_w}. \quad (38)$$

The following lemmas collect the closed-form expressions for the above second-order statistics.

Lemma 5: For the two-tap ISI channel, $\mathbf{R}_{yy}^{(w)} = \text{tridiag}(R_{-1}, R_0, R_1)$.

Proof: See Appendix C-A. $\square$

Lemma 6: Following the convention that the column index $\ell = 0, 1, \dots, N_w - 1$ corresponds to $y_{n-\ell}$, the cross-covariance matrix $\mathbf{R}_{ry}$ is given columnwise by

$$[\mathbf{R}_{ry}]_{i,0} = R_0 + R_{-1} \beta_i, \quad (39)$$

$$[\mathbf{R}_{ry}]_{i,\ell} = \beta_i^{\ell-1} (R_1 + \beta_i R_0 + \beta_i^2 R_{-1}), \quad \ell \geq 1. \quad (40)$$

Proof: See Appendix C-B. $\square$

Lemma 7: The cross-covariance of the windowed input with the desired transmit symbol is given by $\mathbf{r}_{yx}^{(w)} = P \mathbf{e}_1$, where $\mathbf{e}_1 \triangleq [1, 0, \dots, 0]^T \in \mathbb{C}^{N_w}$.

Proof: See Appendix C-C. $\square$

Define the covariance matrix of the WESN observation vector $\mathbf{s}_n$ as

$$\mathbf{R}_{ss} \triangleq \mathbb{E} [\mathbf{s}_n \mathbf{s}_n^H] = \begin{bmatrix} \mathbf{R}_{rr} & \mathbf{R}_{ry} \\ \mathbf{R}_{yr} & \mathbf{R}_{yy} \end{bmatrix} \in \mathbb{C}^{(M+N_w) \times (M+N_w)}. \quad (41)$$

Define the cross-covariance vector of the WESN observation vector with the transmit symbol as

$$\mathbf{r}_{sx} \triangleq \mathbb{E} [\mathbf{s}_n x_n^*] = P \begin{bmatrix} \mathbf{1}_M \\ \mathbf{e}_1 \end{bmatrix} \in \mathbb{C}^{M+N_w}. \quad (42)$$

Then, the optimum output (readout) weights for the WESN obtained by solving the Normal equations is given by

$$\mathbf{w}_{\text{out,opt}} = \mathbf{R}_{ss}^{-1} \mathbf{r}_{sx}, \quad (43)$$

and the corresponding minimum MSE achieved is given by

$$\text{MSE}_{\text{WESN}} = P - \mathbf{r}_{sx}^H \mathbf{R}_{ss}^{-1} \mathbf{r}_{sx}. \quad (44)$$

Define the FIR Toeplitz block matrix as $\mathbf{T} \triangleq \mathbf{R}_{yy}^{(w)} > 0$ and its inverse action on $\mathbf{e}_1$ as $\mathbf{z} \triangleq \mathbf{T}^{-1} \mathbf{e}_1$. Furthermore, introduce the Schur complement as

$$\mathbf{S} \triangleq \mathbf{R}_{rr} - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{R}_{yr} > 0, \quad (45)$$

$$\boldsymbol{\delta} \triangleq \mathbf{1} - \mathbf{R}_{ry} \mathbf{z}. \quad (46)$$

We also define

$$\tilde{\mathbf{S}} \triangleq \mathbf{T} - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} > 0, \quad (47)$$

$$\tilde{\boldsymbol{\delta}} \triangleq \mathbf{e}_1 - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{1}. \quad (48)$$

Remark 4: Positive Definiteness of $\mathbf{T}$ and $\mathbf{S}$.

- Since $\mathbf{T}$ is the covariance matrix of $\mathbf{y}_n^{(w)}$, $\mathbf{T} \geq 0$ by definition. Additionally, for $N_0 > 0$, the stricter property $\mathbf{T} > 0$ holds.
- Consequently, the Schur complement $\mathbf{S} > 0$ because $\mathbf{R}_{ss} > 0$.

We state the MSE and SINR achieved by the WESN in the following theorem.

Theorem 2: For the two-tap ISI channel of Eq. (1), the minimum post-equalization MSE achieved by the WESN is given by

$$\text{MSE}_{\text{WESN}} = P - P^2 (\mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \delta^H \mathbf{S}^{-1} \delta). \quad (49)$$

Alternatively, it is also expressed as

$$\text{MSE}_{\text{WESN}} = P - P^2 (\mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1} + \tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta}). \quad (50)$$

Proof: The complete proof is provided in Appendix C-D. $\square$

Remark 5: The WESN “gain” term $\xi_{\text{WESN}} \triangleq \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \delta^H \mathbf{S}^{-1} \delta = \mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1} + \tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta}$ can be interpreted in two ways as described below.

- In the representation given by $\xi_{\text{WESN}} = \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \delta^H \mathbf{S}^{-1} \delta$, the term $\mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1$ represents the FIR gain, while the term $\delta^H \mathbf{S}^{-1} \delta$ represents the residual gain of the reservoir that is not explained by the FIR window. We call this the “FIR-centric” representation.
- Symmetrically, in the representation given by $\xi_{\text{WESN}} = \mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1} + \tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta}$, the term $\mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1}$ represents the ESN/reservoir gain while the term $\tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta}$ represents the residual gain of the FIR window that is not explained by the reservoir. We call this the “ESN-centric” representation.

This clear separation is useful in attributing the role of either component of the WESN, the reservoir and the FIR window towards reduction of the overall MSE.

Rank-reduced MSE Expressions for the WESN: It is helpful to derive equivalent expressions for MSE_{WESN} for lower complexity computation by exploiting the embedded reduced rank forms.

Recall from the vanilla ESN derivation that $\mathbf{R}_{rr} = \mathbf{R}_0 + \mathbf{UCU}^H$, where $\mathbf{R}_0 \triangleq \mathbf{A}\mathbf{R}_{zz}\mathbf{A}^H + \mathbf{C}_{\text{tail}}$. With $\mathbf{T} > 0$ fixed, form the *base matrix* $\mathbf{B}$ as

$$\mathbf{B} \triangleq \mathbf{R}_0 - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{R}_{yr}, \quad (51)$$

$$\Rightarrow \mathbf{S} = \underbrace{\mathbf{B}}_{\text{base}} + \underbrace{\mathbf{UCU}^H}_{\text{rank-2 update}}. \quad (52)$$

The rank-2 resummed WESN MSE expressions are then stated in the following corollary.

Corollary 2: For $\mathbf{q} \triangleq \mathbf{B}^{-1} \delta$, $\mathbf{y}_B \triangleq \mathbf{U}^H \mathbf{q}$, $\mathbf{G}_B \triangleq \mathbf{U}^H \mathbf{B}^{-1} \mathbf{U}$, $\mathbf{M}_B \triangleq \mathbf{C}^{-1} + \mathbf{G}_B$, the MSE achieved by the WESN can be decomposed as

$$\begin{aligned} \text{MSE}_{\text{WESN}} &= P - P^2 \left(\underbrace{\mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1}_{\text{FIR gain}} + \underbrace{\delta^H \mathbf{B}^{-1} \delta}_{\text{base ESN residual}} - \underbrace{\mathbf{y}_B^H \mathbf{M}_B^{-1} \mathbf{y}_B}_{\text{rank-2 ESN correction}} \right). \end{aligned} \quad (53)$$

Equivalently, it can also be written as

$$\begin{aligned} \text{MSE}_{\text{WESN}} &= P - P^2 \left(\underbrace{\mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M}_{\text{ESN gain}} + \underbrace{\tilde{\delta}^H \tilde{\mathbf{B}}^{-1} \tilde{\delta}}_{\text{base FIR residual}} - \underbrace{\tilde{\mathbf{y}}^H (\mathbf{M} + \tilde{\mathbf{G}})^{-1} \tilde{\mathbf{y}}}_{\text{rank-2 FIR correction}} \right) \end{aligned} \quad (54)$$

Proof: See Appendix C-E. $\square$

Finally, we note from our derivations in Sec. III-C that the MSE achieved by a N_w -tap FIR-only LMMSE equalizer is simply given by

$$\text{MSE}_{\text{FIR}} = P - P^2 \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1. \quad (55)$$

D. MSE Ordering per Channel Realization

Theorem 3: For the fixed fading channel realization α and any stable pole set $\{\beta_i\}_{i=1}^M$ with $M \geq 0$, window length $N_w \geq 1$, the following holds

$$\text{MSE}_{\text{Wiener}} \leq \text{MSE}_{\text{WESN}} \leq \min\{\text{MSE}_{\text{ESN}}, \text{MSE}_{\text{FIR}}\}. \quad (56)$$

Proof: First, we prove that $\text{MSE}_{\text{Wiener}} \leq \text{MSE}_{\text{WESN}}$. Let $\mathcal{H}$ be the Hilbert space of zero-mean square-integrable complex random variables with inner product $\langle a, b \rangle = \mathbb{E}[ab^*]$. For fixed n , set $X \triangleq x_n$. Next, define the linear subspaces

$$\mathcal{S}_{\text{WESN}} \triangleq \text{span} \{ \mathbf{1}_M^H \mathbf{r}_n, \mathbf{e}_1^H \mathbf{y}^{(w)} \}, \quad (57)$$

$$\mathcal{S}_{\text{Wiener}} \triangleq \text{span} \{ y_{n-k} : k \in \mathbb{Z} \}. \quad (58)$$

Each $r_{n,i}$ is a causal LTI functional of $\{y_{n-k}\}_{k \geq 0}$, therefore it follows that $\mathcal{S}_{\text{WESN}} \subseteq \mathcal{S}_{\text{Wiener}}$. The LMMSE estimates $\hat{x}_{n,\text{Wiener}}$ and $\hat{x}_{n,\text{WESN}}$ obtained by both the Wiener and the WESN equalizers respectively are the orthogonal projections of X onto their respective subspaces. For any two linear subspaces U, V satisfying $U \subseteq V$ with corresponding projection error norms ε_U and ε_V respectively, we have $\varepsilon_V \leq \varepsilon_U$. Therefore, it follows that

$$\begin{aligned} \text{MSE}_{\text{Wiener}} &= \min_{Z \in \mathcal{S}_{\text{Wiener}}} \mathbb{E}[|X - Z|^2] \leq \min_{Z \in \mathcal{S}_{\text{WESN}}} \mathbb{E}[|X - Z|^2] \\ &= \text{MSE}_{\text{WESN}}. \end{aligned} \quad (59)$$

Furthermore, strict inequality holds above unless $\mathcal{S}_{\text{WESN}}$ already contains the Wiener solution, which is a non-generic event for finite M and N_w . This proves the first inequality in (56).

To prove the second inequality in (56), recall from Eq. (49) that

$$\frac{P - \text{MSE}_{\text{WESN}}}{P^2} = \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \delta^H \mathbf{S}^{-1} \delta. \quad (60)$$

Recognizing that $\mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 = \frac{P - \text{MSE}_{\text{FIR}}}{P^2}$, it follows that

$$\text{MSE}_{\text{FIR}} - \text{MSE}_{\text{WESN}} = P^2 \delta^H \mathbf{S}^{-1} \delta \geq 0, \quad (61)$$

with equality iff $\delta = \mathbf{0}$. Similarly, recognizing that $\mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1} = \frac{P - \text{MSE}_{\text{ESN}}}{P^2}$, it follows from Eq. (50) that

$$\text{MSE}_{\text{ESN}} - \text{MSE}_{\text{WESN}} = P^2 \tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta} \geq 0, \quad (62)$$

with equality iff $\tilde{\delta} = \mathbf{0}$. It is instructive to examine the equality conditions more closely. Specifically, we have $\text{MSE}_{\text{WESN}} = \text{MSE}_{\text{FIR}}$ iff $\delta = \mathbf{0}$, i.e., $\mathbf{1}_M = \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{e}_1$, implying that the

reservoir's x -correlated component is completely explained by the FIR window. Similarly, we have $\text{MSE}_{\text{WESN}} = \text{MSE}_{\text{ESN}}$ iff $\tilde{\mathbf{d}} = \mathbf{0}$, i.e., $\mathbf{e}_1 = \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{1}_M$, implying that the FIR window adds no new x -correlated component beyond the ESN states. Both of the above are measure-zero algebraic coincidences for finite M , N_w and generic α , and therefore, the inequalities are typically strict. $\square$

We now state a general “feature augmentation” lemma which will be useful in establishing the monotonicity of MSE_{WESN} w.r.t. M and N_w .

Lemma 8: Let $\mathbf{f} \in \mathbb{C}^p$ be an observation vector for estimating x and let $\mathbf{g} \in \mathbb{C}^q$ be the **additional** (stacked) features. Defining the second-order statistics as $\mathbf{K}_{ff} \triangleq \mathbb{E}[\mathbf{f}\mathbf{f}^H]$, $\mathbf{K}_{gg} \triangleq \mathbb{E}[\mathbf{g}\mathbf{g}^H]$, $\mathbf{K}_{fg} \triangleq \mathbb{E}[\mathbf{f}\mathbf{g}^H]$, $\mathbf{k}_{fx} \triangleq \mathbb{E}[\mathbf{f}x^*]$ and $\mathbf{k}_{gx} \triangleq \mathbb{E}[\mathbf{g}x^*]$. With the MSE being achieved using the features $\mathbf{f}$ being $\text{MSE}(\mathbf{f}) = P - \mathbf{k}_{fx}^H \mathbf{K}_{ff}^{-1} \mathbf{k}_{fx}$ and the conditional covariance of $\mathbf{g}$ given $\mathbf{f}$ and the residual cross-correlation with the target respectively as $\Sigma_{glf} \triangleq \mathbf{K}_{gg} - \mathbf{K}_{gf} \mathbf{K}_{ff}^{-1} \mathbf{K}_{fg} \geq \mathbf{0}$ and $\zeta_{glf} \triangleq \mathbf{k}_{gx} - \mathbf{K}_{gf} \mathbf{K}_{ff}^{-1} \mathbf{k}_{fx}$. Then, the MSE achieved by the “augmented” feature vector $[\mathbf{f}^T, \mathbf{g}^T]^T \in \mathbb{C}^{p+q}$ satisfies

$$\text{MSE}(\mathbf{f}, \mathbf{g}) = \text{MSE}(\mathbf{f}) - \zeta_{glf}^H \Sigma_{glf}^{-1} \zeta_{glf}, \quad (63)$$

resulting in $\text{MSE}(\mathbf{f}, \mathbf{g}) \leq \text{MSE}(\mathbf{f})$ with equality iff $\zeta_{glf} = \mathbf{0}$.

Proof: Form the stacked covariance matrix and cross-covariance vector according to

$$\mathbf{K}_{zz} = \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fg} \\ \mathbf{K}_{gf} & \mathbf{K}_{gg} \end{bmatrix}, \quad \mathbf{k}_{zx} = \begin{bmatrix} \mathbf{k}_{fx} \\ \mathbf{k}_{gx} \end{bmatrix}. \quad (64)$$

Then, it follows that

$$\text{MSE}(\mathbf{f}, \mathbf{g}) = P - \mathbf{k}_{zx}^H \mathbf{K}_{zz}^{-1} \mathbf{k}_{zx}. \quad (65)$$

The inverse $\mathbf{K}_{zz}^{-1}$ can be obtained via the block matrix identity via the Schur complement of $\mathbf{K}_{gg}$ as

$$\mathbf{K}_{zz}^{-1} = \begin{bmatrix} \mathbf{K}_{ff}^{-1} + \mathbf{K}_{ff}^{-1} \mathbf{K}_{fg} \Sigma_{glf}^{-1} \mathbf{K}_{gf} \mathbf{K}_{ff}^{-1} & -\mathbf{K}_{ff}^{-1} \mathbf{K}_{fg} \Sigma_{glf}^{-1} \\ -\Sigma_{glf}^{-1} \mathbf{K}_{gf} \mathbf{K}_{ff}^{-1} & \Sigma_{glf}^{-1} \end{bmatrix}. \quad (66)$$

Left- and right-multiplying the LHS above with $\mathbf{k}_{zx}^H$ and $\mathbf{k}_{zx}$ respectively followed by some elementary algebra, we get

$$\mathbf{k}_{zx}^H \mathbf{K}_{zz}^{-1} \mathbf{k}_{zx} = \mathbf{k}_{fx}^H \mathbf{K}_{ff}^{-1} \mathbf{k}_{fx} + \zeta_{glf}^H \Sigma_{glf}^{-1} \zeta_{glf}. \quad (67)$$

Substituting Eq. (67) in Eq. (65), we arrive at Eq. (63), thus proving the lemma. $\square$

Remark 6: The result of Lemma 8 can be interpreted as follows: The only way additional features can fail to improve the MSE is when they bring no residual correlation with the target beyond what the existing features already explain. Otherwise, the reduction in MSE is a strictly positive quadratic form.

Corollary 3: MSE_{WESN} is a non-decreasing function in both M and N_w . Specifically, for a fixed channel realization α and noise variance $N_0 > 0$, for any $M_2 \geq M_1$ and any $N_{w,2} \geq N_{w,1}$,

$$\text{MSE}_{\text{WESN}}(M_2, N_w) \leq \text{MSE}_{\text{WESN}}(M_2, N_w), \quad (68)$$

$$\text{MSE}_{\text{WESN}}(M, N_{w,2}) \leq \text{MSE}_{\text{WESN}}(M, N_{w,1}). \quad (69)$$

Proof: First, we prove the case of increasing M for a fixed N_w . Let

$$\mathbf{f} \triangleq \phi^{(M_1, N_w)} = \begin{bmatrix} \mathbf{r}^{(M_1)} \\ \mathbf{y}^{(N_w)} \end{bmatrix}, \quad \mathbf{g} \triangleq \mathbf{r}^{(\Delta M)},$$

$$\Delta M \triangleq M_2 - M_1 \geq 0. \quad (70)$$

Then, the augmented feature vector is $[\mathbf{f}^T, \mathbf{g}^T]^T = \phi^{(M_2, N_w)}$. Applying Lemma 8 with these definitions of $\mathbf{f}$ and $\mathbf{g}$, we get

$$\begin{aligned} \text{MSE}_{\text{WESN}}(M_2, N_w) &= \text{MSE}_{\text{WESN}}(M_1, N_w) - \zeta_{glf}^H \Sigma_{glf}^{-1} \zeta_{glf} \\ &\leq \text{MSE}_{\text{WESN}}(M_1, N_w), \end{aligned} \quad (71)$$

with equality iff $\zeta_{glf} = \mathbf{0}$, i.e., the additional ΔM reservoir neurons bring no residual correlation with x_n beyond what $\phi^{(M_1, N_w)}$ already contains.

Next, we prove the case of increasing N_w for fixed M . For this case, let

$$\mathbf{f} \triangleq \phi^{(M, N_{w,1})} = \begin{bmatrix} \mathbf{r}^{(M)} \\ \mathbf{y}^{(N_{w,1})} \end{bmatrix}, \quad \mathbf{g} \triangleq \mathbf{y}^{(N_{w,2} \setminus N_{w,1})}. \quad (72)$$

Applying Lemma 8 again with these modified definitions of $\mathbf{f}$ and $\mathbf{g}$ gives

$$\begin{aligned} \text{MSE}_{\text{WESN}}(M, N_{w,2}) &= \text{MSE}_{\text{WESN}}(M, N_{w,1}) - \zeta_{glf}^H \Sigma_{glf}^{-1} \zeta_{glf} \\ &\leq \text{MSE}_{\text{WESN}}(M, N_{w,1}), \end{aligned} \quad (73)$$

with equality iff the newly added FIR window taps carry no residual correlation with x_n after conditioning on the existing features. $\square$

Theorem 4: There does not exist a per-realization universal ordering between MSE_{ESN} for an M -neuron vanilla ESN and MSE_{FIR} for an N_w -tap FIR-only LMMSE equalizer. Specifically, for the two-tap ISI channel model of Eq. (1) and fixed finite integers $M \geq 1$ and $N_w \geq 1$, there exist parameter choices (channel realization α , SNR γ , and ESN pole set $\{\beta_i\}$) such that

$$\text{MSE}_{\text{ESN}}(M, \{\beta_i\}) < \text{MSE}_{\text{FIR}}(N_w), \quad (74)$$

and there exist (potentially different) parameter choices such that

$$\text{MSE}_{\text{FIR}}(N_w) < \text{MSE}_{\text{ESN}}(M, \{\beta_i\}). \quad (75)$$

Proof: We provide two constructive cases that exhibit strict inequalities in opposite directions.

Case A (FIR can strictly beat ESN): For any $u = |\alpha|^2 > 0$, any SNR $\gamma > 0$, consider a one-neuron ESN ($M = 1$) with its pole $\beta_1 = 0$ (or more generally, consider an M -neuron ESN with $M > 1$ and all M poles equal to 0 so that the reservoir state reduces to the single regressor y_n). Then, we have

$$\text{MSE}_{\text{ESN}}(M = 1, \beta_1 = 0) = \text{MSE}_{\text{FIR}}(N_w = 1). \quad (76)$$

For any $N_w \geq 2$, we have

$$\begin{aligned} \text{MSE}_{\text{FIR}}(N_w \geq 2) &< \text{MSE}_{\text{FIR}}(N_w = 1) \\ &= \text{MSE}_{\text{ESN}}(M = 1, \beta_1 = 0). \end{aligned} \quad (77)$$

The above holds because with $\beta_1 = 0$, the ESN state is $r_{n,1} = y_n$, so that the ESN feature space is $\text{span}\{y_n\}$. For $N_w \geq 2$, the ESN feature space is $\text{span}\{y_n, y_{n-1}, \dots, y_{n-N_w+1}\}$, which strictly contains $\text{span}\{y_n\}$, and therefore, the LMMSE projection error cannot increase. Hence, $\text{MSE}_{\text{FIR}}(N_w)$ strictly decreases with N_w .

Case B: ESN can strictly beat FIR: Assume a minimum-phase channel, i.e., $|\alpha| < 1$. Fix any finite window $N_w \geq 1$ and any $u \in (0, 1)$. Consider the one-neuron ESN ($M = 1$)

with $\beta_1 = \alpha$ and consider the readout equation to be $\hat{x}_n = r_{n,1}$ (i.e., output/readout weight = 1). Then,

$$r_{n,1} = \sum_{k \geq 0} \alpha^k y_{n-k} = \underbrace{x_n}_{\text{ISI perfectly canceled}} + \underbrace{\sum_{k \geq 0} \alpha^k w_{n-k}}_{\text{colored noise}}. \quad (78)$$

Therefore,

$$\begin{aligned} \text{MSE}_{\text{ESN}}(M=1, \beta_1 = \alpha) &\leq \mathbb{E} \left[\left| \sum_{k \geq 0} \alpha^k w_{n-k} \right|^2 \right] = \frac{N_0}{1-u} \\ &= \frac{P}{\gamma} \cdot \frac{1}{1-u}. \end{aligned} \quad (79)$$

The above inequality holds because the perfectly trained output weight (via the Normal equations) gives an MSE which is no worse than the choice of the output weight being 1. On the other hand, the N_w -tap FIR LMMSE equalizer has a strictly positive high-SNR MSE floor given by

$$\lim_{\gamma \rightarrow \infty} \text{MSE}_{\text{FIR}}(N_w) = P \frac{u^{N_w}(1-u)}{1-u^{N_w+1}} \triangleq P \Phi(u, N_w) > 0. \quad (80)$$

Hence, for any fixed $u \in (0, 1)$ and finite N_w , choosing

$$\gamma > \gamma^*(u, N_w) \triangleq \frac{1}{(1-u)\Phi(u, N_w)} \quad (81)$$

ensures that

$$\begin{aligned} \text{MSE}_{\text{ESN}}(M=1, \beta_1 = \alpha) &\leq \frac{P}{\gamma(1-u)} < P \Phi(u, N_w) \\ &\leq \text{MSE}_{\text{FIR}}(N_w), \end{aligned} \quad (82)$$

i.e., the ESN is strictly better than the FIR. Thus, Case A gives parameters where $\text{MSE}_{\text{FIR}} < \text{MSE}_{\text{ESN}}$, while Case B gives parameters where $\text{MSE}_{\text{ESN}} < \text{MSE}_{\text{FIR}}$. Hence, no universal ordering exists. $\square$

IV. AN INFORMATION-THEORETIC VIEWPOINT

In this section, we provide an alternative perspective on the exact MSE analysis conducted in Sec. III. Specifically, we use an information theoretic lens to analyze the MSE performance of each equalizer analyzed previously. Leveraging the connections between mutual information (MI) and mean square error (MMSE) for linear estimators under Gaussian signaling, we arrive at the same conclusions as those reached using exact estimation-theoretic analysis. But first, a geometric interpretation of the WESN gain ξ_{WESN} is developed.

A. Geometric Interpretation of the WESN Gain

The two derived identities for the WESN gain ξ_{WESN} have a precise geometric interpretation in a suitable whitened (orthonormal) feature space, where the two addends in each identity are the squared lengths of orthogonal components of the same vector. We provide two equivalent interpretations below, using the shorthand $\mathbf{r}$ to denote $\mathbf{r}_n$.

FIR-centric orthogonalization: This orthogonalization process follows the steps below:

- 1) Whiten the window block $\mathbf{y}^{(w)}$ via $\mathbf{T}^{-1}$ to get $\tilde{\mathbf{y}} \triangleq \mathbf{T}^{-1/2} \mathbf{y}^{(w)}$, with covariance matrix $\mathbf{I}_{N_w}$.

- 2) Remove the linear prediction by the window from the reservoir according to

$$\mathbf{r}^\perp \triangleq \mathbf{r} - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{y}^{(w)}, \quad (83)$$

$$\tilde{\mathbf{r}}^\perp \triangleq \mathbf{S}^{-1/2} \mathbf{r}^\perp. \quad (84)$$

Then, we have $\mathbb{E}[\tilde{\mathbf{r}}^\perp \tilde{\mathbf{r}}^{\perp H}] = \mathbf{0}_{M \times N_w}$.

- 3) Whitened Feature Space: Define the stacked orthonormal feature vector as

$$\tilde{\mathbf{s}} \triangleq \begin{bmatrix} \tilde{\mathbf{y}} \\ \tilde{\mathbf{r}}^\perp \end{bmatrix} \in \mathbb{C}^{M+N_w}, \quad (85)$$

such that $\mathbb{E}[\tilde{\mathbf{s}} \tilde{\mathbf{s}}^H] = \mathbf{I}_{M+N_w}$.

- 4) Using the relation $\mathbb{E}[\mathbf{y}^{(w)} x_n^*] = P \mathbf{e}_1$ and $\mathbb{E}[\mathbf{r}^\perp x_n^*] = P(\mathbf{I} - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{e}_1) = P \boldsymbol{\delta}$, the whitened cross-covariance vector is

$$\mathbf{g}_{\text{FIR}} \triangleq \frac{\mathbb{E}[\tilde{\mathbf{s}} x_n^*]}{P} = \begin{bmatrix} \mathbf{T}^{-1/2} \mathbf{e}_1 \\ \mathbf{S}^{-1/2} \boldsymbol{\delta} \end{bmatrix}. \quad (86)$$

- 5) Write the squared length of this vector according to

$$\begin{aligned} \|\mathbf{g}_{\text{FIR}}\|^2 &= \|\mathbf{T}^{-1/2} \mathbf{e}_1\|^2 + \|\mathbf{S}^{-1/2} \boldsymbol{\delta}\|^2 \\ &= \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \boldsymbol{\delta}^H \mathbf{S}^{-1} \boldsymbol{\delta}, \end{aligned} \quad (87)$$

which is the same as the WESN gain $\xi_{\text{WESN}} = (P - \text{MSE}_{\text{WESN}})/P^2$.

Thus, in the FIR-centric representation, the first axis span $(\mathbf{T}^{-1/2} \mathbf{y}^{(w)})$ representing the window subspace and the second axis span $(\mathbf{S}^{-1/2} \mathbf{r}^\perp)$ representing the reservoir residual subspace are orthogonal in the whitened feature space.

ESN-centric orthogonalization: This orthogonalization process follows the steps below:

- 1) Whiten the reservoir block according to $\tilde{\mathbf{r}} \triangleq \mathbf{R}_{rr}^{-1/2} \mathbf{r}$, which satisfies $\mathbb{E}[\tilde{\mathbf{r}} \tilde{\mathbf{r}}^H] = \mathbf{I}_M$.
- 2) Remove the linear prediction by the reservoir from the window according to

$$\mathbf{y}^\perp \triangleq \mathbf{y}^{(w)} - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{r}, \quad (88)$$

and define $\tilde{\mathbf{y}}^\perp \triangleq \tilde{\mathbf{S}}^{-1/2} \mathbf{y}^\perp$, with $\tilde{\mathbf{S}} = \mathbf{T} - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} > 0$ and $\mathbb{E}[\tilde{\mathbf{y}}^\perp \tilde{\mathbf{y}}^{\perp H}] = \mathbf{0}$.

- 3) Whitened Feature Space: Define the stacked orthonormal feature vector as

$$\tilde{\mathbf{s}}_{\text{ESN}} \triangleq \begin{bmatrix} \tilde{\mathbf{r}} \\ \tilde{\mathbf{y}}^\perp \end{bmatrix} \in \mathbb{C}^{M+N_w}. \quad (89)$$

- 4) The whitened cross-covariance vector can be obtained to the FIR-centric case as

$$\mathbf{g}_{\text{ESN}} \triangleq \frac{\mathbb{E}[\tilde{\mathbf{s}}_{\text{ESN}} x_n^*]}{P} = \begin{bmatrix} \mathbf{R}_{rr}^{-1/2} \mathbf{1}_M \\ \tilde{\mathbf{S}}^{-1/2} \tilde{\boldsymbol{\delta}} \end{bmatrix}, \quad (90)$$

where $\tilde{\boldsymbol{\delta}} \triangleq \mathbf{e}_1 - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{1}_M$.

- 5) Write the squared length of this vector as

$$\begin{aligned} \|\mathbf{g}_{\text{ESN}}\|^2 &= \|\mathbf{R}_{rr}^{-1/2} \mathbf{1}_M\|^2 + \|\tilde{\mathbf{S}}^{-1/2} \tilde{\boldsymbol{\delta}}\|^2 \\ &= \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + \tilde{\boldsymbol{\delta}}^H \tilde{\mathbf{S}}^{-1} \tilde{\boldsymbol{\delta}}, \end{aligned} \quad (91)$$

which is the equivalent representation of the WESN gain ξ_{WESN} .

Thus, in the ESN-centric representation, the first axis span $(\mathbf{R}_{rr}^{-1/2} \mathbf{r})$ representing the reservoir subspace and the

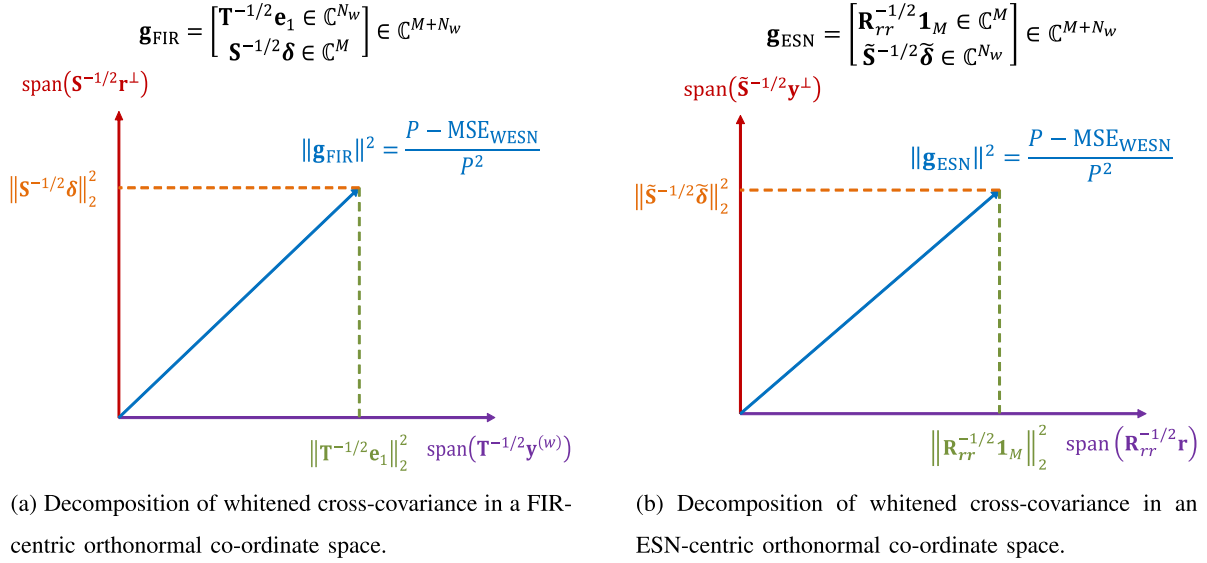


Fig. 1. A feature space view of the WESN in two equivalent representations: FIR-centric and ESN-centric.

second axis $\text{span}(\tilde{\mathbf{S}}^{-1/2} \mathbf{y}^\perp)$ representing the window residual subspace are orthogonal in this alternate but equivalent whitened feature space.

The dual and equivalent orthogonalization procedures described above are illustrated in Fig. 1.

B. An Information Theoretic Perspective

Under the system model of Eq. (1), define $X \triangleq x_n$ and $S \in \{\mathbf{s}_n^{(\text{ESN})}, \mathbf{s}_n^{(\text{FIR})}, \mathbf{s}_n^{(\text{WESN})}\}$. Since S is a linear transformation of a Gaussian process, (X, S) is jointly proper-complex Gaussian distributed. Leveraging the I-MMSE relationship introduced in [29], for the scalar proper complex Gaussian random variable X , it can be derived that

$$I(X; S) = \log \frac{P}{\text{mmse}(X|S)}, \quad (92)$$

where $I(A_1; A_2)$ denotes the mutual information of the two random variables A_1 and A_2 . Now, for any two observation sets S_1 and S_2 , it holds from the data processing inequality (DPI) that

$$I(X; [S_1, S_2]) = I(X; S_1) + I(X; S_2|S_1) \geq I(X; S_1), \quad (93)$$

with the symmetric inequality holding by interchanging S_1 with S_2 . Defining $S_1 \triangleq \mathbf{s}_n^{(\text{ESN})}$ and $S_2 \triangleq \mathbf{s}_n^{(\text{FIR})}$, it follows from the above DPI that

$$I(X; \mathbf{s}_n^{(\text{WESN})}) \geq \max \{I(X; \mathbf{s}_n^{(\text{ESN})}), I(X; \mathbf{s}_n^{(\text{FIR})})\}. \quad (94)$$

This mutual information (MI) ordering relation can be translated into an MSE ordering relation using the identity in Eq. (92) as follows:

$$\text{MSE}_{\text{WESN}} = P e^{-I(X; \mathbf{s}_n^{(\text{WESN})})} \leq P e^{-I(X; \mathbf{s}_n^{(\text{ESN})})} = \text{MSE}_{\text{ESN}}, \quad (95)$$

$$\text{MSE}_{\text{WESN}} = P e^{-I(X; \mathbf{s}_n^{(\text{WESN})})} \leq P e^{-I(X; \mathbf{s}_n^{(\text{FIR})})} = \text{MSE}_{\text{FIR}}. \quad (96)$$

Furthermore, strict inequality holds above iff the conditional MI terms are positive, i.e., if $I(X; \mathbf{s}_n^{(\text{FIR})} | \mathbf{s}_n^{(\text{ESN})}) > 0$ or $I(X; \mathbf{s}_n^{(\text{ESN})} | \mathbf{s}_n^{(\text{FIR})}) > 0$, implying that the additional features

contain new information about X beyond the information already contained in the original feature set. Finally, under Gaussian signaling in AWGN where the MMSE estimator is also linear, equality in the above conditions translate exactly to the linear dependence relations arrived at in the exact estimation-theoretic analysis of Sec. III, i.e.,

$$\boldsymbol{\delta} = \mathbf{0} \Leftrightarrow I(X; [\mathbf{s}_n^{(\text{ESN})}, \mathbf{s}_n^{(\text{FIR})}]) = I(X; \mathbf{s}_n^{(\text{FIR})}), \quad (97)$$

$$\tilde{\boldsymbol{\delta}} = \mathbf{0} \Leftrightarrow I(X; [\mathbf{s}_n^{(\text{ESN})}, \mathbf{s}_n^{(\text{FIR})}]) = I(X; \mathbf{s}_n^{(\text{ESN})}). \quad (98)$$

Therefore, the above information-theoretic analysis arrives at the same conclusion as that reached using the exact estimation-theoretic MSE analysis. This lays further credence to the overall validity of the theoretical results and its potential use in future work in optimizing ESN/WESN pole placement as well as other extensions.

C. Analysis Roadmap for General L -Tap ISI Channels With Gaussian Inputs

In this section, we provide a high-level overview of how the analysis presented in previous sections for two-tap ISI channels with Gaussian inputs can be generalized for a general L -tap ISI channel. While the detailed L -tap channel analysis is left to future work, we describe how the tools employed in this work can be utilized. Describing the channel model as

$$y_n = \sum_{\ell=0}^{L-1} h_\ell x_{n-\ell} + w_n, \quad (99)$$

where $x_n \sim \mathcal{CN}(0, P)$, $w_n \sim \mathcal{CN}(0, N_0)$ and $x \perp w$. The channel impulse response is $\mathbf{h}_{\text{ch}} \triangleq [h_0, h_1, \dots, h_{L-1}]^\top$. The key second-order quantity is the received autocorrelation sequence

$$R_{yy}[m] = P \sum_{\ell=0}^{L-1} h_\ell h_{\ell-m}^* + N_0 \delta[m], \quad (100)$$

where $h_k = 0$ outside $k \in \{0, \dots, L-1\}$. Hence, $R_{yy}[m] = 0$ for $|m| \geq L$. This sequence replaces the special two-tap

quantities R_0 , $R_{\pm 1}$ and drives all subsequent derivations in the L -tap channel case. Furthermore, with the normalization $h_0 = 1$ without loss of generality, many of the cross-covariance formulas look exactly like the two-tap case. We provide short formulas for the MSE incurred by each linear equalizer with accompanying remarks as follows:

- $\text{MSE}_{\text{FIR}}(N_w) = P - P^2 \mathbf{e}_1^H \mathbf{T}_{N_w}^{-1} \mathbf{e}_1$. Thus, the formula generalizes readily except $\mathbf{T}_{N_w}$ is no longer tridiagonal but rather banded Toeplitz.
- $\text{MSE}_{\text{ESN}} = P - P^2 \mathbf{1}^H \mathbf{R}_{rr}^{-1} \mathbf{1}$. Again, the formula retains its structure, however $\mathbf{R}_{rr}$ has a more complicated structure with each entry $(\mathbf{R}_{rr})_{ij}$ becoming a finite linear combination of geometric sums.
- $\text{MSE}_{\text{WESN}} = P - \mathbf{r}_{zx}^H \mathbf{R}_{zz}^{-1} \mathbf{r}_{zx}$. In the joint covariance matrix $\mathbf{R}_{zz}$ which has a block structure with $\mathbf{R}_{ry}$ as one of the block entries. Its entries $(\mathbf{R}_{ry})_{ij}$ reduce to finite combinations of geometric sums.
- $\text{MSE}_{\text{Wiener}}$ for $L = 2$ had an integral expression which could be evaluated to a closed-form expression. However, for $L > 2$, this integral needs to be evaluated numerically.

In general, for ISI channels with $L > 2$ taps, the MSE ordering conclusion from $L = 2$ still holds, i.e., $\text{MSE}_{\text{WESN}} \leq \max\{\text{MSE}_{\text{FIR}}, \text{MSE}_{\text{ESN}}\}$.

D. Impact of Non-Gaussian Input Distribution With $L = 2$ Tap ISI Channel

We also briefly discuss how the results derived in previous sections change if the input x_n is no longer Gaussian. As an example, consider a practical non-Gaussian input such as equiprobable QAM or PSK, where x_n is still i.i.d. across n . Then, $\mathbb{E}[x_n] = 0$, $\mathbb{E}[|x_n|^2] = P$ and $\mathbb{E}[x_n x_m^*] = P\delta[n - m]$. Thus, the input is non-Gaussian but has the same second-order statistics as the Gaussian input considered previously. First, note that all of the previous LMMSE derivations for the FIR, ESN, WESN and Wiener front-ends only considered the second-order statistics of the input and the noise, and the linear channel model. Therefore, if the input is still zero-mean, white across time and has the same variance P , then all the linear covariance calculations remain the same. Consequently, the MSE expressions for the “best” linear front-ends still stay the same. However, under non-Gaussian inputs, the optimum posterior mean estimator is no longer linear, and therefore, the LMMSE front-ends (FIR, ESN, WESN) derived earlier are no longer the true MMSE-achieving receivers. The optimal Bayesian estimator expressions for each of the considered linear front-ends can be written explicitly or approximately. Similarly, the global optimal receiver is also no longer the Wiener filter, but a nonlinear trellis-based Bayesian estimator/detector. Then the true MMSE expressions for the considered linear front-ends can also be written or numerically evaluated. Similarly, from an information-theoretic standpoint, the I-MMSE identity can still be leveraged for an arbitrary input distribution with additive Gaussian noise. However, now the linear front-end outputs cannot be reduced to an equivalent scalar AWGN sufficient statistic, so that the expressions for mutual information $I(\cdot; \cdot)$ and the MSE are no longer as simple as the Gaussian input case. However, this exact

information-theoretic analysis can still be made tractable and is in fact, the subject of our forthcoming work.

E. Complexity Analysis

In this section, we perform complexity analysis of the linear ESN and WESN front-ends. To make the complexity analysis practical, we consider finite training data availability. We denote by T the number of samples used for training and by S the number of samples used for testing or inference. Thus, we first perform data-driven training of the linear readout/tap weights from a finite training set of size T of labeled pilot samples and then perform inference on S received samples. For FIR, ESN and WESN, the output weights are trained via batch least squares. Specifically, if $d \in \{d_{\text{FIR}}, d_{\text{ESN}}, d_{\text{WESN}}\}$ with $d_{\text{FIR}} = N_w$, $d_{\text{ESN}} = M$, $d_{\text{WESN}} = M + N_w$, and $\mathbf{Z}_{tr} \in \mathbb{C}^{T \times d}$ stacks the training feature vectors rowwise and $\mathbf{x}_{tr} \in \mathbb{C}^T$ stacks the targets, then the batch LS/LMMSE readout equation is $\hat{\mathbf{w}} = (\mathbf{Z}_{tr}^H \mathbf{Z}_{tr})^{-1} \mathbf{Z}_{tr}^H \mathbf{x}_{tr}$. In summary, the batch training complexity for the FIR is $\mathcal{O}(TN_w^2 + N_w^3)$, for the ESN is $\mathcal{O}(TM^2 + M^3)$ and for the WESN is $\mathcal{O}((T(M + N_w)^2 + (M + N_w)^3))$. The inference complexity over S samples are $\mathcal{O}(SN_w)$, $\mathcal{O}(SM)$ and $\mathcal{O}(S(M + N_w))$ for FIR, ESN and WESN respectively. For the optimum Wiener filter, assuming (α, P, N_0) are known, there is no “training” involved and the inference complexity is $\mathcal{O}(S)$. On the other hand, if these are unknown, then the batch training complexity for the Wiener filter is $\mathcal{O}(T)$, which is required for parameter estimation of α , P and N_0 .

F. Applicability to OFDM Systems

The previously MSE ordering relations and conclusions derived for the two-tap ISI channel still hold for an OFDM system with cyclic prefix (CP) length greater than 1. The only difference is that instead of the analysis corresponding to an infinite stationary linear convolution operating on a continuous stream of received symbols convolved by the channel, we must now operate on the “CP-removed” block of time-domain received symbols of finite length N_{sc} , where N_{sc} denotes the number of subcarriers of the OFDM system.

V. NUMERICAL EVALUATIONS

A. Validation of Theoretical Results

To validate the correctness of the theoretical MSE expressions derived for each linear receive processing approach considered for channel equalization, we conduct empirical Monte Carlo numerical evaluations under the following conditions. The transmitted sequence consists of $N_{\text{sym}} = 10^5$ complex Gaussian symbols with unit average power $P = 1$. The signal-to-noise ratio (SNR) γ is varied from 0 dB to 30 dB in steps of 5 dB, where the noise variance is defined as $N_0 = P/\gamma$ with γ denoting the linear SNR. For each SNR level, results are averaged over 50 independent realizations of the channel.

The following equalization approaches are evaluated:

- 1) The global optimum Wiener equalizer,
- 2) The vanilla echo state network (ESN) with reservoir sizes $M \in \{2, 4, 8\}$,

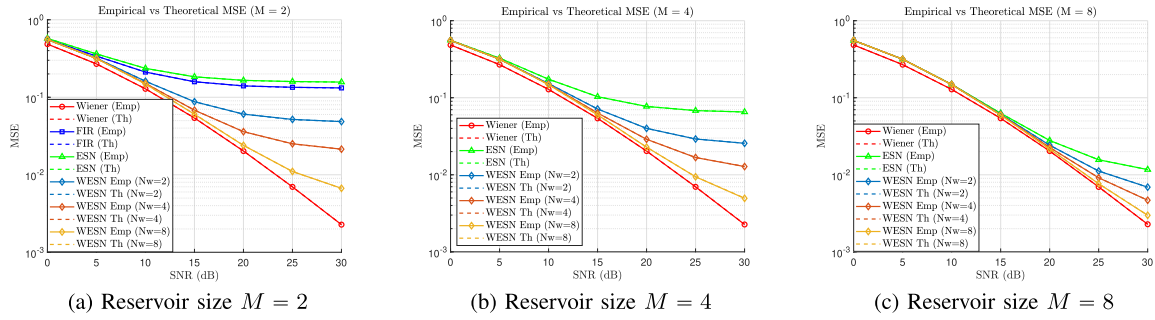


Fig. 2. MSE comparison of different detectors with uniformly distributed coefficients for Pole Set A.

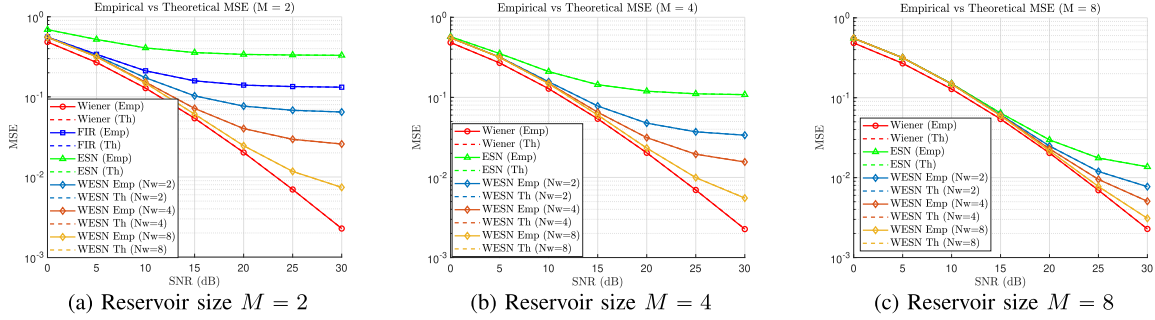
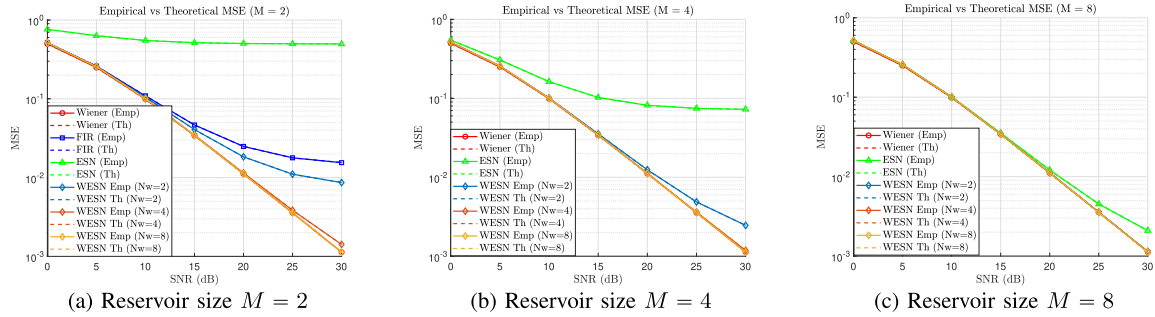


Fig. 3. MSE comparison of different detectors with uniformly distributed coefficients for Pole Set B.

Fig. 4. MSE performance of different detectors under complex Gaussian channel coefficients with variance $\sigma^2 = 0.1$.

- 3) The windowed ESN (WESN) with reservoir sizes $M \in \{2, 4, 8\}$ and window lengths $N_w \in \{2, 4, 8\}$.

For $M = 2$, the ESN and the WESN are also compared against the two-tap ($N_w = 2$) FIR LMMSE equalizer. To validate the theoretical analysis, additional simulations are performed with different channel coefficients α . The channel tap α is generated either from a complex uniform distribution constrained to a circle of radius $r = 0.9$, or from a complex Gaussian distribution with variances $\sigma^2 \in \{0.1, 0.5, 0.9\}$. Both the empirical mean squared error (MSE) from simulations and the theoretical expressions derived in Sec. III are reported in this study. All results are presented on a logarithmic scale as a function of SNR.

First, we evaluate the performance of the linear equalizers for uniformly distributed channel coefficients. For both ESN and WESN, the same set of reservoir poles are fixed once and reused across all SNR values and channel realizations. To examine the sensitivity to pole placement, two distinct pole sets are considered. In the case of a two-pole ESN, the additional poles for $M > 2$ are placed uniformly on a circle of radius $r = 0.6$. The two-pole sets are given as follows:

- **Pole Set A:** $\beta_1 = 0.5e^{-j0}$, $\beta_2 = -0.4e^{j\pi/32}$.

- **Pole Set B:** $\beta_1 = 0.7e^{-j0}$, $\beta_2 = -0.8e^{j\pi/32}$.

Fig. 2 and Fig. 3 demonstrate that the Wiener filter provides the best performance overall, as expected. The MSE achieved by both the ESN as well as the WESN approaches that of the Wiener filter as M increases. For the two-pole case, however, the placement of poles critically influences the ESN equalizer behavior: with Set A (Fig. 2), the ESN demonstrate better performance than with Set B poles (Fig. 3). In contrast, WESN performance is less sensitive to pole placement. These results highlight that ESN performance is strongly pole-dependent, whereas WESN offers more robustness against pole placement variations. Nonetheless, these empirical evaluations demonstrate the critical importance of “optimizing” the choice of the poles in both the ESN as well as the WESN, or equivalently optimizing the distribution from which the poles are drawn from independently, as illustrated in our previous work [30].

The above results highlight that equalization performance is strongly influenced by the distribution of the channel fading coefficient α . To validate the theoretical derivations, we tested the equalization approaches under complex Gaussian-distributed α with multiple variances.

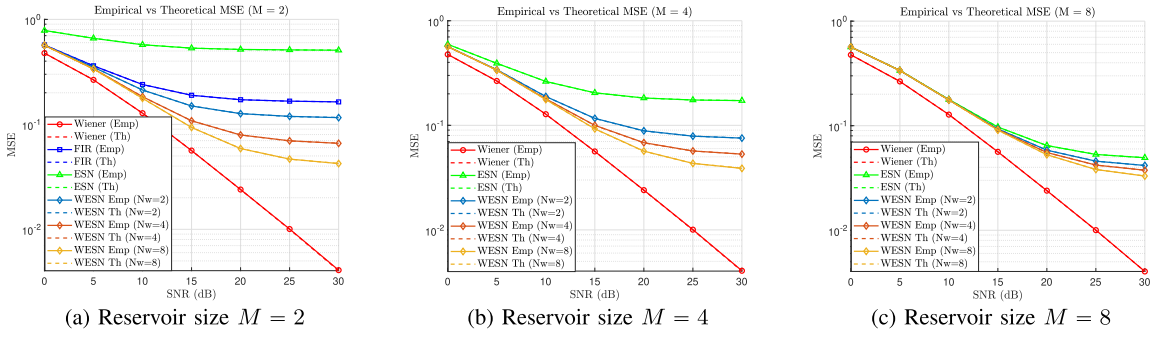


Fig. 5. MSE performance of different detectors under complex Gaussian channel coefficients with variance $\sigma^2 = 0.5$.

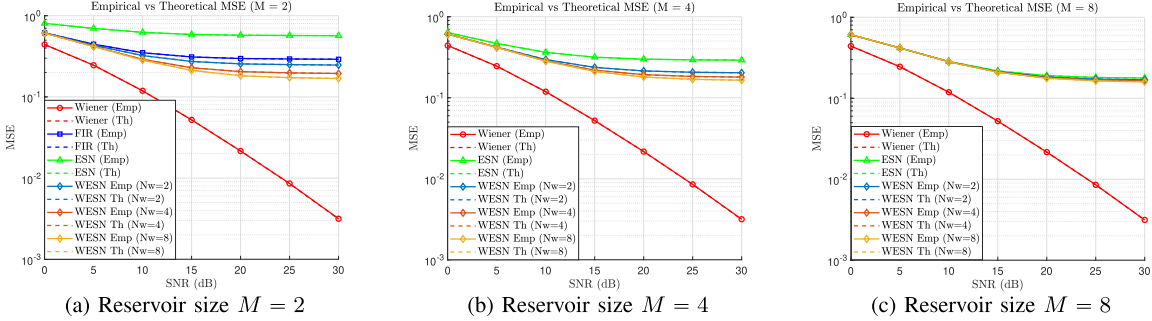


Fig. 6. MSE performance of different detectors under complex Gaussian channel coefficients with variance $\sigma^2 = 0.9$.

Fig. 4 shows that for $\sigma^2 = 0.1$, the WESN consistently outperforms the ESN, and its performance improves as both the reservoir size and window length N_w increase. In particular, the WESN with eight neurons achieves performance comparable to that of the Wiener filter. In Fig. 5 and Fig. 6, WESN maintains a performance advantage over ESN, but the gap to the Wiener filter remains. For $\sigma^2 = 0.9$, the performance degradation is more pronounced compared to $\sigma^2 = 0.5$. Even with larger reservoirs and extended window sizes, the improvement saturates, leading to a noticeable MSE floor.

Across all tested distributions of channel coefficients, the simulated MSE trends are consistent with the derived theoretical results, thereby confirming the correctness of the analytical expressions derived in Sec. III.

VI. CONCLUSION AND FUTURE WORK

In this work, we developed a rigorous analytical framework to understand the equalization performance of reservoir computing (RC)-based methods in the physical layer of wireless communications. Focusing on the echo state network (ESN) and its extension, the windowed ESN (WESN), we provided an exact characterization of their capabilities in mitigating inter-symbol interference (ISI) caused by a two-tap wireless fading channel and benchmarked their performance against the optimum Wiener equalizer. This analysis advances the explainability of RC-based architectures by connecting their operation to established estimation- and information-theoretic principles, thereby offering a clearer understanding from an alternate perspective of their observed empirical advantages in OFDM and OTFS systems. As part of future work, the exact equalization error characterization for general higher-order fading channels under linear activation will be considered. In addition, non-Gaussian inputs with the same linear front-ends

will also be considered. Furthermore, the impact of nonlinear activation as well as a rigorous analysis of the choice of the reservoir pole distribution under signaling from a discrete constellation (QAM/PSK) will also be investigated.

APPENDIX A GLOBAL NON-CAUSAL WIENER EQUALIZER

A. Proof of Lemma 1

Let $X(e^{j\omega})$, $Y(e^{j\omega})$ and $W(e^{j\omega})$ represent the DTFTs of x_n , y_n and w_n respectively. Then, we have $Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) + W(e^{j\omega})$. Independence of $X(e^{j\omega})$ and $W(e^{j\omega})$ implies that $X(e^{j\omega}) \perp W(e^{j\omega})$. By definition of the power spectral density,

$$\begin{aligned} S_{yy}(\omega) &= \lim_{N \rightarrow \infty} \frac{1}{2\pi} \mathbb{E} \left[\frac{1}{N} Y(e^{j\omega}) Y^*(e^{j\omega}) \right] \\ &= |H(e^{j\omega})|^2 S_{xx}(\omega) + S_{ww}(\omega) = P|H(e^{j\omega})|^2 + N_0. \end{aligned} \quad (101)$$

Similarly,

$$\begin{aligned} S_{xy}(\omega) &= \mathbb{E} [X(e^{j\omega}) Y^*(e^{j\omega})] \\ &= \mathbb{E} [X(e^{j\omega}) (H X + W)^*(e^{j\omega})] = H^*(e^{j\omega}) S_{xx}(\omega) \\ &= P H^*(e^{j\omega}). \end{aligned} \quad (102)$$

B. Proof of Lemma 2

Consider the frequency domain representation of the equalized signal $\hat{X}(e^{j\omega})$ under a general equalizer $G(e^{j\omega})$. Then, $\hat{X}(e^{j\omega}) = G(e^{j\omega})Y(e^{j\omega})$ and $E(e^{j\omega}) = X(e^{j\omega}) - \hat{X}(e^{j\omega})$ represents the error spectrum. Then, the corresponding power spectral density is (dropping the parentheses on the RHS for brevity)

$$S_{ee}(\omega) = \mathbb{E} [|E(e^{j\omega})|^2] = S_{xx} - G S_{xy} - G^* S_{yx} + |G|^2 S_{yy},$$

$$= \left| G - \frac{S_{xy}}{S_{yy}} \right|^2 S_{yy} + \left(S_{xx} - \frac{|S_{xy}|^2}{S_{yy}} \right). \quad (103)$$

The above quadratic form of $S_{ee}(\omega)$ is minimized by $G_{\text{opt}}(e^{j\omega}) = \frac{S_{xy}}{S_{yy}}$, with the minimum being

$$S_{ee}^{(\min)}(\omega) = S_{xx}(\omega) - \frac{|S_{xy}(\omega)|^2}{S_{yy}(\omega)}. \quad (104)$$

This completes the proof of Lemma 2.

C. Closed-Form of Trigonometric Integral

With $A \triangleq P(1+u) + N_0$ and $B \triangleq 2P\sqrt{u}$, we have $A - |B| = P(1+u-2\sqrt{u}) + N_0 = P(\sqrt{u}-1)^2 + N_0 > 0$. Thus, $A > |B|$. Consider the integral

$$\mathcal{I}(A, B) \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d\theta}{A - B \cos \theta}, \quad A > |B| > 0. \quad (105)$$

Set $z = e^{j\theta}$, leading to $d\theta = dz/(jz)$ and $\cos \theta = \frac{1}{2}(z + z^{-1})$. Then,

$$\begin{aligned} \mathcal{I}(A, B) &= \frac{1}{2\pi j} \oint_{|z|=1} \frac{dz}{A - \frac{B}{2}(z + z^{-1})} \cdot \frac{1}{z} \\ &= \frac{1}{2\pi j} \oint_{|z|=1} \frac{dz}{p(z)}, \end{aligned} \quad (106)$$

where $p(z) = -\frac{B}{2}z^2 + Az - \frac{B}{2}$. Its roots are

$$\begin{aligned} z_{\pm} &= \frac{A \pm \sqrt{A^2 - B^2}}{B}, \\ 0 &< \frac{A - \sqrt{A^2 - B^2}}{B} < 1 < \frac{A + \sqrt{A^2 - B^2}}{B}. \end{aligned} \quad (107)$$

Thus, only the root z_- lies inside $|z| = 1$. Using Cauchy's residue theorem,

$$\mathcal{I}(A, B) = \frac{1}{2\pi j} \cdot 2\pi j \cdot \text{Res} \left(\frac{1}{p(z)}, z_- \right) = \frac{1}{p'(z_-)}. \quad (108)$$

Since $p'(z) = -Bz + A$, we have

$$p'(z_-) = -B \frac{A - \sqrt{A^2 - B^2}}{B} + A = \sqrt{A^2 - B^2}. \quad (109)$$

Therefore, $\mathcal{I}(A, B) = \frac{1}{\sqrt{A^2 - B^2}}$, yielding the closed-form of the desired integral.

APPENDIX B

M-NEURON VANILLA ESN

A. Proof of Proposition 1

Recall that

$$[\mathbf{R}_{rr}]_{ij} = \frac{R_0 + R_1\beta_j^* + R_{-1}\beta_i}{1 - \beta_i\beta_j^*} \triangleq \frac{N_{ij}}{1 - t_{ij}}, \quad (110)$$

where $N_{ij} \triangleq R_0 + R_1\beta_j^* + R_{-1}\beta_i$ and $t_{ij} \triangleq \beta_i\beta_j^*$. Using the identity

$$\frac{1}{1-t} = 1 + t + t^2 \frac{1}{1-t}, \quad (111)$$

we can write

$$\frac{N_{ij}}{1 - t_{ij}} = N_{ij}(1 + t_{ij}) + t_{ij}^2 \frac{N_{ij}}{1 - t_{ij}}. \quad (112)$$

Expanding $N_{ij}(1 + t_{ij}) = R_0(1 + t_{ij}) + R_1\beta_j^* + R_{-1}\beta_i + t_{ij}(R_1\beta_j^* + R_{-1}\beta_i)$, we can regroup the terms as shown below

$$\begin{aligned} \frac{N_{ij}}{1 - t_{ij}} &= \underbrace{R_0(1 + t_{ij}) + R_1\beta_j^* + R_{-1}\beta_i}_{[\mathbf{AR}_{zz}\mathbf{A}^H]_{ij}} + \underbrace{t_{ij}(R_1\beta_j^* + R_{-1}\beta_i)}_{[\mathbf{UCU}^H]_{ij}} \\ &\quad + \underbrace{t_{ij}^2 \frac{N_{ij}}{1 - t_{ij}}}_{[\mathbf{C}_{\text{tail}}]_{ij}}, \end{aligned} \quad (113)$$

where the RHS is exactly $[\mathbf{R}_0]_{ij} + [\mathbf{UCU}^H]_{ij}$. Thus, we have shown that the following relation holds elementwise, i.e., $[\mathbf{R}_{rr}]_{ij} = [\mathbf{R}_0]_{ij} + [\mathbf{UCC}^H]_{ij}$, thereby proving Proposition 1.

APPENDIX C

M-NEURON, N_w -WINDOW WINDOWED ESN (WESN)

A. Proof of Lemma 5

Using the definition of $\mathbf{R}_{yy}^{(w)}$, its (ℓ, m) -th entry (for $0 \leq \ell, m \leq (N_w - 1)$) can be written as

$$\begin{aligned} [\mathbf{R}_{yy}^{(w)}]_{\ell, m} &= \mathbb{E}[y_{n-\ell}y_{n-m}^*] = R_{yy}[m - \ell] \\ &= \begin{cases} R_0, & m = \ell \\ R_1, & m = \ell + 1 \\ R_{-1}, & m = \ell - 1 \\ 0, & |m - \ell| \geq 2. \end{cases} \end{aligned} \quad (114)$$

Thus, following the symmetry in Eq. (114), we can deduce that $\mathbf{R}_{yy}^{(w)}$ is Hermitian tridiagonal Toeplitz with diagonal R_0 , superdiagonal R_1 and subdiagonal R_{-1} .

B. Proof of Lemma 6

For column index $\ell \in \{0, 1, \dots, N_w - 1\}$ in $\mathbf{R}_{ry}$, we have

$$\begin{aligned} \mathbb{E}[r_{n,i}y_n^*] &= \sum_{k \geq 0} \beta_i^k R_{yy}[\ell - k] \\ &= \underbrace{\beta_i^\ell R_0 + \beta_i^{\ell+1} R_{-1}}_{k=\ell, \ell+1} + \underbrace{\mathbb{1}_{\ell \geq 1} \beta_i^{\ell-1} R_1}_{k=\ell-1}. \end{aligned} \quad (115)$$

Thus, for column $\ell = 0$, $[\mathbf{R}_{ry}]_{i,0} = R_0 + \beta_i R_{-1}$. For columns $\ell \geq 1$, the only non-zero lags in R_{yy} produce

$$\begin{aligned} [\mathbf{R}_{ry}]_{i,\ell} &= \beta_i^\ell R_0 + \beta_i^{\ell-1} R_1 + \beta_i^{\ell+1} R_{-1} \\ &= \beta_i^{\ell-1} (R_1 + \beta_i R_0 + \beta_i^2 R_{-1}), \quad \ell \geq 1. \end{aligned} \quad (116)$$

From the above, we get the complete structure of $\mathbf{R}_{ry}$, proving Lemma 6. Note the symmetry in the above expressions resulting in $\mathbf{R}_{ry}$ being Hermitian symmetric, i.e., $\mathbf{R}_{ry} = \mathbf{R}_{ry}^H$.

C. Proof of Lemma 7

Recall that $\mathbf{r}_{yx}^{(w)} \triangleq \mathbb{E}[\mathbf{y}_n^{(w)} \mathbf{x}_n^*] \in \mathbb{C}^{N_w}$, where $\mathbf{y}_n^{(w)} = [y_n, y_{n-1}, \dots, y_{n-N_w+1}]^T$ collects a causal window of N_w consecutive received samples. For $\ell = 0, 1, \dots, N_w - 1$, the $(\ell + 1)$ -th element of $\mathbf{r}_{yx}^{(w)}$ is $[\mathbf{r}_{yx}^{(w)}]_{\ell+1} = \mathbb{E}[y_{n-\ell}x_n^*] = \mathbb{E}[(x_{n-\ell} - \alpha x_{n-\ell-1} + w_{n-\ell})x_n^*]$. For $\ell = 0$, we have $\mathbb{E}[y_n x_n^*] = \mathbb{E}[x_n x_n^*] - \alpha \mathbb{E}[x_{n-1} x_n^*] + \mathbb{E}[w_n x_n^*] = P$. Similarly for $\ell \geq 1$, $\mathbb{E}[y_{n-\ell} x_n^*] = \mathbb{E}[x_{n-\ell} x_n^*] - \alpha \mathbb{E}[x_{n-\ell-1} x_n^*] + \mathbb{E}[w_{n-\ell} x_n^*] = 0$. The simplifications in both of the above expressions follow from Assumption 1 and 2. Collecting all the terms, we arrive at $\mathbf{r}_{yx}^{(w)} = P\mathbf{e}_1$, thus proving Lemma 7.

D. Proof of Theorem 2

Start with the block matrix form of $\mathbf{R}_{ss} \triangleq \mathbb{E}[\mathbf{s}_n \mathbf{s}_n^H]$ given by

$$\mathbf{R}_{ss} = \begin{bmatrix} \mathbf{R}_{rr} & \mathbf{R}_{ry} \\ \mathbf{R}_{yr} & \mathbf{T} \end{bmatrix}. \quad (117)$$

Using the LMMSE trained output (readout) weights for the WESN with perfect knowledge of second-order statistics, the WESN MSE is $\text{MSE}_{\text{WESN}} = P - \mathbf{r}_{sx}^H \mathbf{R}_{ss}^{-1} \mathbf{r}_{sx} = P - P^2 \xi_{\text{WESN}}$, where ξ_{WESN} is the WESN “gain” term defined in Remark 5 and $\mathbf{r}_{sx} \triangleq \mathbb{E}[\mathbf{s}_n x_n^*] = P[\mathbf{1}_M^H, \mathbf{e}_1^H]^H$. Defining $\mathbf{S} \triangleq \mathbf{R}_{rr} - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{R}_{yr}$ and applying the block inversion (Schur complement) identity for $\mathbf{R}_{ss}$ with $\mathbf{T} > 0$, we get

$$\mathbf{R}_{ss}^{-1} = \begin{bmatrix} \mathbf{S}^{-1} & -\mathbf{S}^{-1} \mathbf{R}_{ry} \mathbf{T}^{-1} \\ -\mathbf{T}^{-1} \mathbf{R}_{yr} \mathbf{S}^{-1} & \mathbf{T}^{-1} + \mathbf{T}^{-1} \mathbf{R}_{yr} \mathbf{S}^{-1} \mathbf{R}_{ry} \mathbf{T}^{-1} \end{bmatrix}. \quad (118)$$

Let $\mathbf{u} \triangleq [\mathbf{1}_M^H, \mathbf{e}_1^H]^H$ and recognize that $\xi_{\text{WESN}} = \mathbf{u}^H \mathbf{R}_{ss}^{-1} \mathbf{u}$. Left- and right-multiplying the above equation by $\mathbf{u}^H$ and $\mathbf{u}$ respectively gives

$$\begin{aligned} \mathbf{u}^H \mathbf{R}_{ss}^{-1} \mathbf{u} &= \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 \\ &+ (\mathbf{1}_M - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{e}_1)^H \mathbf{S}^{-1} (\mathbf{1}_M - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{e}_1). \end{aligned} \quad (119)$$

Defining $\delta \triangleq \mathbf{1}_M - \mathbf{R}_{ry} \mathbf{z}$ with $\mathbf{z} \triangleq \mathbf{T}^{-1} \mathbf{e}_1$, we obtain the FIR-centric representation of the WESN gain as $\xi_{\text{WESN}} = \mathbf{e}_1^H \mathbf{T}^{-1} \mathbf{e}_1 + \delta^H \mathbf{S}^{-1} \delta$, which subsequently gives the FIR-centric representation of MSE_{WESN} in Eq. (49).

Next, we proceed to proving the ESN-centric representation of MSE_{WESN} in Eq. (50). Define the ESN-centric residual and the ESN-side Schur complement respectively as

$$\tilde{\delta} \triangleq \mathbf{e}_1 - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{1}_{N_w} \in \mathbb{C}^{N_w}, \quad (120)$$

$$\tilde{\mathbf{S}} \triangleq \mathbf{T} - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} \in \mathbb{C}^{N_w \times N_w} > 0. \quad (121)$$

Applying the block inversion identity using the Schur complement of $\mathbf{R}_{rr}$ in $\mathbf{R}_{ss}$, we get

$$\mathbf{R}_{ss}^{-1} = \begin{bmatrix} \mathbf{R}_{rr}^{-1} + \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} \tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} & -\mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} \tilde{\mathbf{S}}^{-1} \\ -\tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} & \tilde{\mathbf{S}}^{-1} \end{bmatrix}. \quad (122)$$

Now, left- and right-multiply the above form of $\mathbf{R}_{ss}^{-1}$ by $\mathbf{u}^H$ and $\mathbf{u}$ respectively to get

$$\begin{aligned} \mathbf{u}^H \mathbf{R}_{ss}^{-1} \mathbf{u} &= \underbrace{\mathbf{1}_M^H (\mathbf{R}_{rr}^{-1} + \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} \tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1}) \mathbf{1}_M}_{(I)} + \underbrace{\mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} \mathbf{e}_1}_{(II)} \\ &\quad - \underbrace{\mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry} \tilde{\mathbf{S}}^{-1} \mathbf{e}_1 - \mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{1}_M}_{(III)}. \end{aligned} \quad (123)$$

Let $\mathbf{a} \triangleq \mathbf{R}_{rr}^{-1} \mathbf{1}_M$. Then,

$$\begin{aligned} (I) &= \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + \mathbf{a}^H \mathbf{R}_{ry} \tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{a} \\ &= \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + (\mathbf{R}_{yr} \mathbf{a})^H \tilde{\mathbf{S}}^{-1} (\mathbf{R}_{yr} \mathbf{a}), \end{aligned} \quad (124)$$

and

$$\begin{aligned} (III) &= (\mathbf{R}_{yr} \mathbf{a})^H \tilde{\mathbf{S}}^{-1} \mathbf{e}_1 + \mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} (\mathbf{R}_{yr} \mathbf{a}) \\ &= 2\Re \left\{ \mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} \mathbf{R}_{yr} \mathbf{a} \right\}. \end{aligned} \quad (125)$$

Therefore,

$$\mathbf{u}^H \mathbf{R}_{ss}^{-1} \mathbf{u} = \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + \mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} \mathbf{e}_1 - \mathbf{e}_1^H \tilde{\mathbf{S}}^{-1} (\mathbf{R}_{yr} \mathbf{a})$$

$$\begin{aligned} &- (\mathbf{R}_{yr} \mathbf{a})^H \tilde{\mathbf{S}}^{-1} \mathbf{e}_1 + (\mathbf{R}_{yr} \mathbf{a})^H \tilde{\mathbf{S}}^{-1} (\mathbf{R}_{yr} \mathbf{a}), \\ &= \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + (\mathbf{e}_1 - \mathbf{R}_{yr} \mathbf{a})^H \tilde{\mathbf{S}}^{-1} (\mathbf{e}_1 - \mathbf{R}_{yr} \mathbf{a}), \\ &= \mathbf{1}_M^H \mathbf{R}_{rr}^{-1} \mathbf{1}_M + \tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta}. \end{aligned} \quad (126)$$

This is the desired ESN-centric representation of the WESN gain ξ_{WESN} in Eq. (50), thus completing the proof.

E. Proof of Corollary 2

From Eq. (52), we know that $\mathbf{S}$ can be written as a rank-2 update of a base matrix $\mathbf{B}$, i.e., $\mathbf{S} = \mathbf{B} + \mathbf{U} \mathbf{C} \mathbf{U}^H$, where $\mathbf{B} \triangleq \mathbf{R}_0 - \mathbf{R}_{ry} \mathbf{T}^{-1} \mathbf{R}_{yr}$ as defined in Eq. (51) and $\mathbf{C} \in \mathbb{C}^{2 \times 2}$ is defined in Eq. (30). Note that for $\alpha \neq 0$, $\mathbf{C}$ is invertible. Applying Woodbury's identity to $\mathbf{S}^{-1}$ gives

$$\mathbf{S}^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1} \mathbf{U} (\mathbf{C}^{-1} + \mathbf{U}^H \mathbf{B}^{-1} \mathbf{U})^{-1} \mathbf{U}^H \mathbf{B}^{-1} \quad (127)$$

Recall the definitions for $\mathbf{y}_B \in \mathbb{C}^2$, $\mathbf{G}_B \in \mathbb{C}^{2 \times 2}$ and $\mathbf{M}_B \in \mathbb{C}^{2 \times 2}$ provided in Corollary 2. Then, left- and right-multiplying Eq. (127) by δ^H and δ respectively, we get

$$\begin{aligned} \delta^H \mathbf{S}^{-1} \delta &= \delta^H \mathbf{B}^{-1} \delta - (\mathbf{U}^H \mathbf{B}^{-1} \delta)^H \mathbf{M}_B^{-1} (\mathbf{U}^H \mathbf{B}^{-1} \delta), \\ \Rightarrow \delta^H \mathbf{S}^{-1} \delta &= \delta^H \mathbf{B}^{-1} \delta - \mathbf{y}_B^H \mathbf{M}_B^{-1} \mathbf{y}_B. \end{aligned} \quad (128)$$

Substituting the above in Eq. (49) gives the rank-2 resummed expression for the FIR-centric MSE form in Eq. (53), completing the first part of the proof.

Next, we prove the rank-2 resummed expression for the ESN-centric form. Note Eq. (31) provides the expression for $\mathbf{R}_{rr}^{-1}$ with $\mathbf{M} \triangleq \mathbf{C}^{-1} + \mathbf{U}^H \mathbf{R}_0^{-1} \mathbf{U}$. Let $\mathbf{q}_0 \triangleq \mathbf{R}_0^{-1} \mathbf{1}_M$, $\mathbf{y} \triangleq \mathbf{U}^H \mathbf{q}_0$, $\mathbf{G} \triangleq \mathbf{U}^H \mathbf{R}_0^{-1} \mathbf{U}$, so that $\mathbf{M} = \mathbf{C}^{-1} + \mathbf{G}$. By definition, $\tilde{\mathbf{S}} = \mathbf{T} - \mathbf{R}_{yr} \mathbf{R}_{rr}^{-1} \mathbf{R}_{ry}$. Inserting the Woodbury inverse formula for $\mathbf{R}_{rr}^{-1}$ and cleaning up the terms, we can write $\tilde{\mathbf{S}} = \tilde{\mathbf{B}} + \tilde{\mathbf{U}} \tilde{\mathbf{C}} \tilde{\mathbf{U}}^H$, where $\tilde{\mathbf{B}} \triangleq \mathbf{T} - \mathbf{R}_{yr} \mathbf{R}_0^{-1} \mathbf{R}_{ry}$, $\tilde{\mathbf{U}} \triangleq \mathbf{R}_{yr} \mathbf{R}_0^{-1} \mathbf{U}$ and $\tilde{\mathbf{C}} \triangleq \mathbf{M}^{-1}$. Thus, $\tilde{\mathbf{S}}$ is a rank-2 update of $\tilde{\mathbf{B}}$. Applying the Woodbury inverse formula to $\tilde{\mathbf{S}}$, we get

$$\tilde{\mathbf{S}}^{-1} = \tilde{\mathbf{B}}^{-1} - \tilde{\mathbf{B}}^{-1} \tilde{\mathbf{U}} (\tilde{\mathbf{C}}^{-1} + \tilde{\mathbf{U}}^H \tilde{\mathbf{B}}^{-1} \tilde{\mathbf{U}})^{-1} \tilde{\mathbf{U}}^H \tilde{\mathbf{B}}^{-1}. \quad (129)$$

Noting that $\tilde{\mathbf{C}}^{-1} = \mathbf{M}$, define $\tilde{\mathbf{y}} \triangleq \tilde{\mathbf{U}}^H \tilde{\mathbf{B}}^{-1} \tilde{\delta}$, $\tilde{\mathbf{G}} \triangleq \tilde{\mathbf{U}}^H \tilde{\mathbf{B}}^{-1} \tilde{\mathbf{U}}$. Then, left- and right-multiplying Eq. (129), we arrive at

$$\tilde{\delta}^H \tilde{\mathbf{S}}^{-1} \tilde{\delta} = \tilde{\delta}^H \tilde{\mathbf{B}}^{-1} \tilde{\delta} - \tilde{\mathbf{y}}^H (\mathbf{M} + \tilde{\mathbf{G}})^{-1} \tilde{\mathbf{y}}. \quad (130)$$

Substituting the above in Eq. (50) gives the rank-2 resummed expression for the ESN-centric MSE form in Eq. (54), completing the proof.

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Shashank Jere received the B.S. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2014, the M.S. degree in electrical engineering from the University of California, Los Angeles, in 2016, and the Ph.D. degree in electrical and computer engineering from Virginia Polytechnic Institute and State University (Virginia Tech) in 2025. His research interests include wireless communications, optimization, deep learning, information theory, and statistical learning theory.



Lizhong Zheng (Fellow, IEEE) received the B.S. and M.S. degrees from Tsinghua University, Beijing, China, in 1994 and 1997, respectively, and the Ph.D. degree from the University of California, Berkeley, CA, USA, in 2002. Since 2002, he has been with the Department of Electrical Engineering and Computer Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA, where he is currently a Professor of electrical engineering and computer sciences. His research interests include information theory, wireless communications, and statistical inference. He was a recipient of the Eli Jury Award from UC Berkeley in 2002, the IEEE Information Theory Society Paper Award in 2003, the NSF CAREER Award in 2004, and the AFOSR Young Investigator Award in 2007.



Ummay Sumaya Khan received the B.Sc. degree in electrical and electronic engineering from Bangladesh University of Engineering and Technology in 2017 and the M.S. degree in electrical and computer engineering from the University of Florida in 2023. She is currently pursuing the Ph.D. degree with Wireless@VT, Bradley Department of Electrical and Computer Engineering, Virginia Tech. Her research interests include wireless communications, machine learning, neuromorphic computing, information theory, and statistical learning theory.



Lingjia Liu (Fellow, IEEE) received the B.S. degree in electronic engineering from Shanghai Jiao Tong University and the Ph.D. degree in electrical and computer engineering from Texas A&M University. He is the Andrew J. Young Professor of electrical and computer engineering with Virginia Tech (VT), where he is also the Co-Director of Wireless@Virginia Tech, a university center conducting research on all aspects of wireless networks. Prior to joining academia, he was a 3GPP Delegate leading Samsung Research America's effort on multiuser MIMO, CoMP, and HetNets in 3GPP LTE/LTE-Advanced standards. His general research interests lie in enabling technologies for 6G networks, including efficient learning for wireless networks, massive MIMO, open-radio access network (Open RAN), and integrated sensing and communications (ISAC). His research received eight best paper awards. He is an IEEE Distinguished Lecturer.